



**COMPUTATIONAL
CLIMATE & OCEAN GROUP**
at UC Davis



AI for Oceans & Marine Systems

How we can gain insight and help build resilience

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Climate Change AI Summer School 2026



Overview

The physical, chemical and biological ocean

What is changing?

Warming, sea level, acidification, marine heatwaves, and the knock-on effects on plankton and fisheries

What data is there understand and predict the ocean?

In-situ observation, remote sensing, and numerical models, and the gaps each leaves

Why ML?

A short introduction and overview

Case studies

Satellite $p\text{CO}_2$, Antarctic plankton, heatwave attribution, and Atlantic ocean circulation

Should it be called planet Ocean?

The ocean covers about 70% of Earth's surface and contains 97% of its water.

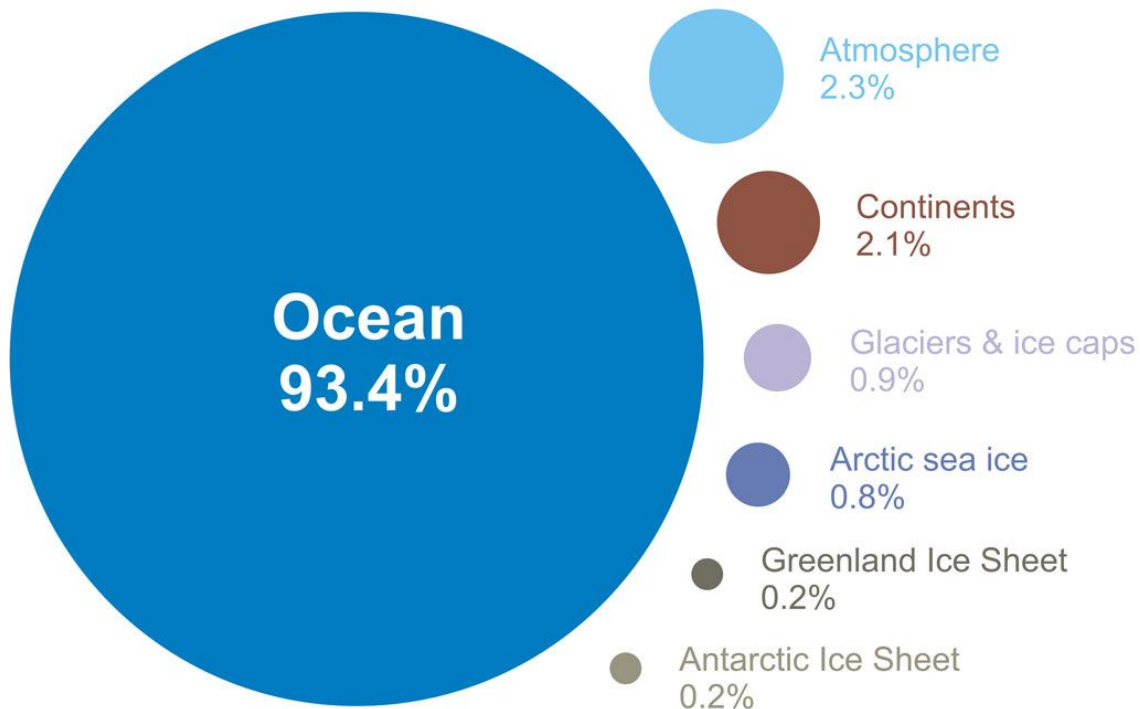


How much of the anthropogenically added heat goes into the ocean?

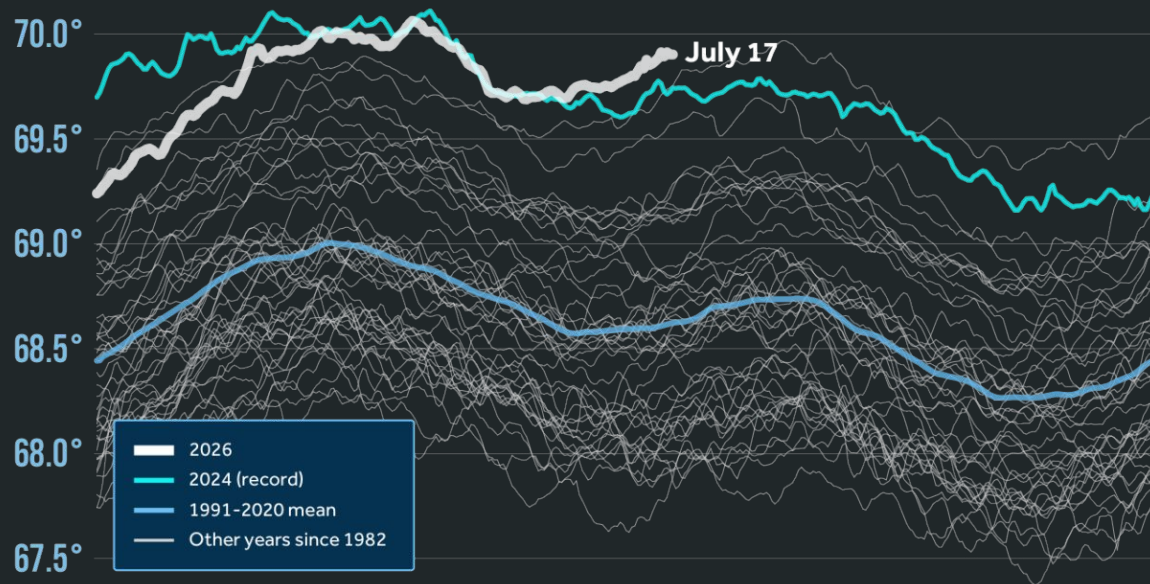


<https://partici.fi/86456312>

Where is global warming going?



Daily global ocean surface temperature (°F)



Data for 2026 includes January 1 through July 17. Source: NOAA OISSTv2.1.

The services the ocean provides



71%

of Earth's surface is ocean



~90%

of global-warming heat is
stored in the ocean



~25%

of our CO₂ emissions
absorbed by the sea



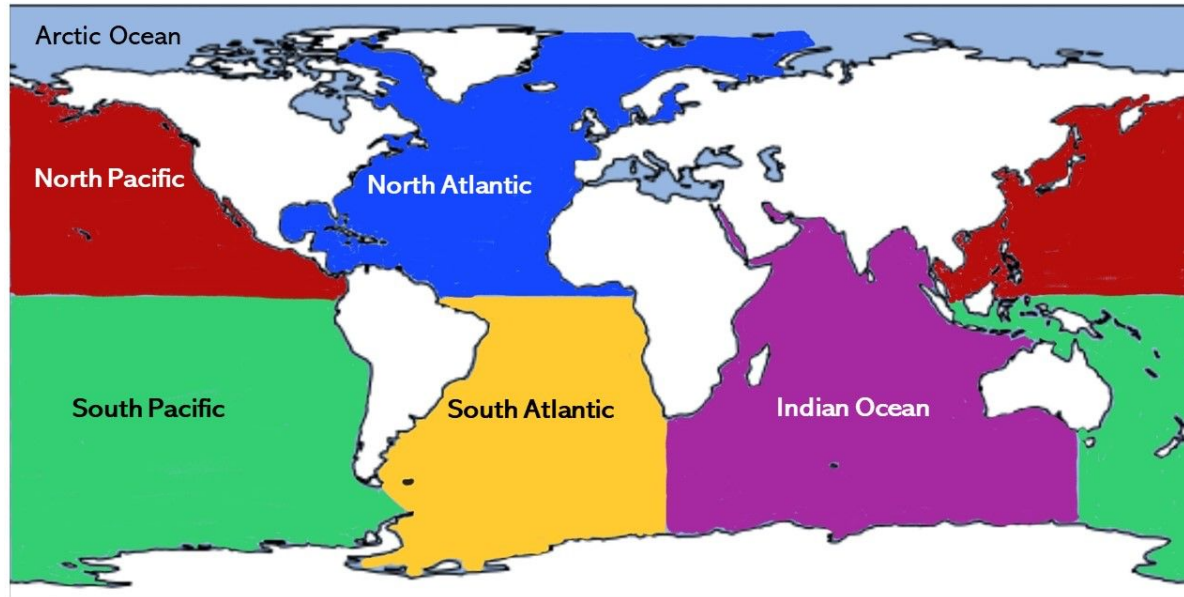
50%+

of the oxygen you breathe
made by ocean plankton

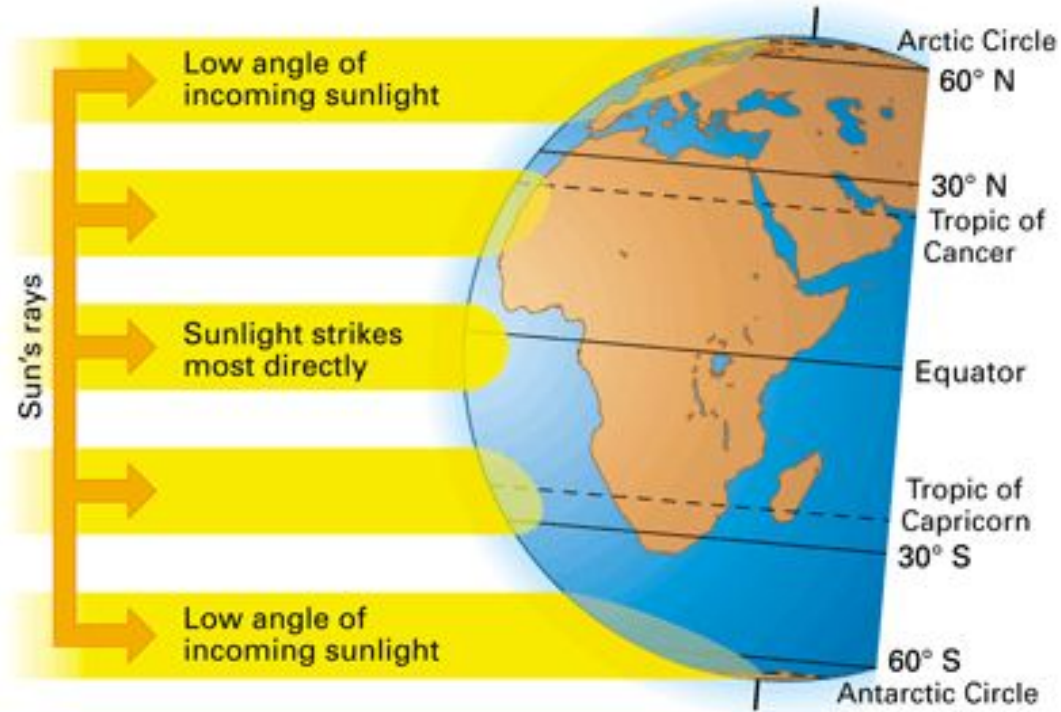
The ocean has different ocean basins of different sizes

Smallest: Arctic 5.4 million sq miles

Largest: Pacific > 60 million sq miles

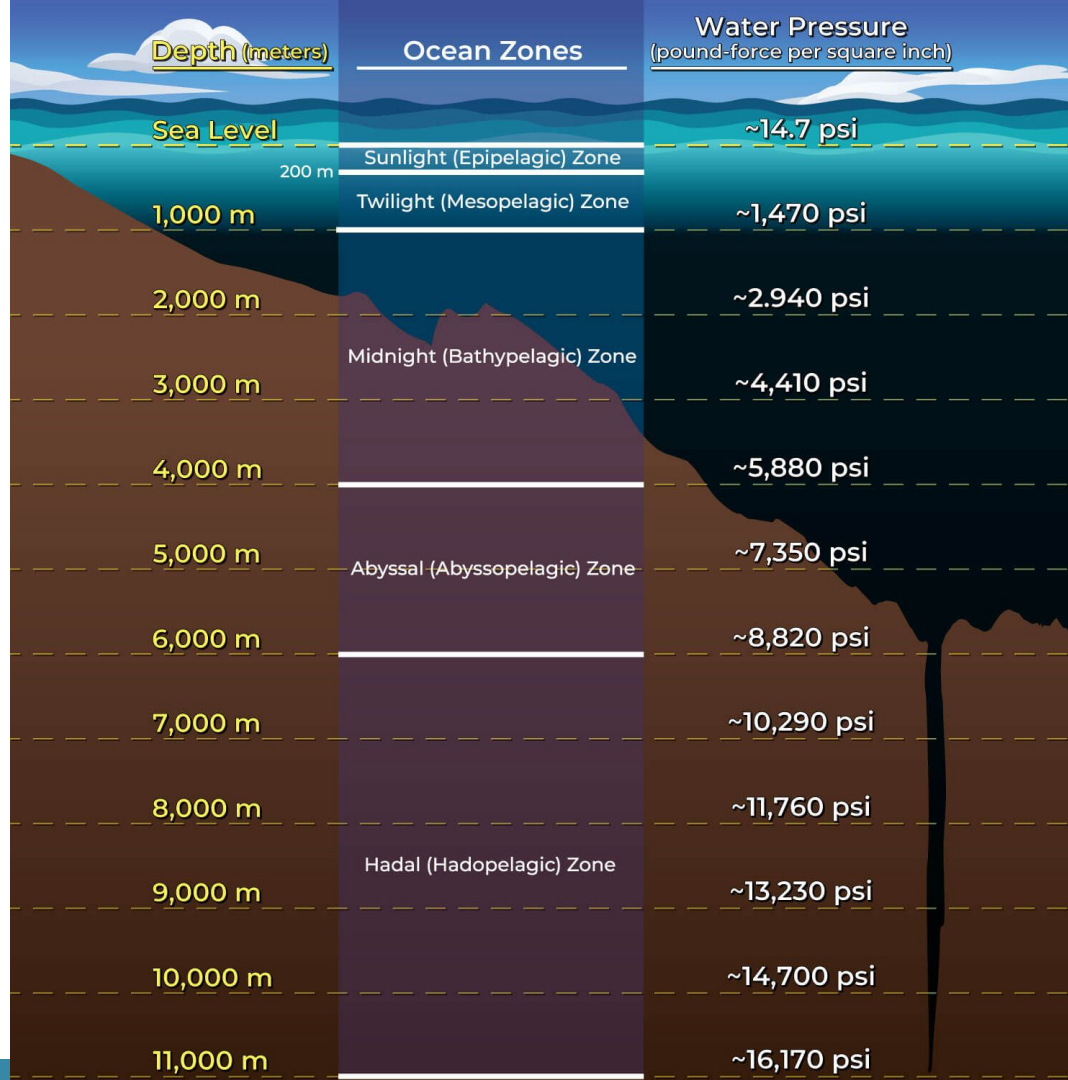


Unequal heating has a big impact

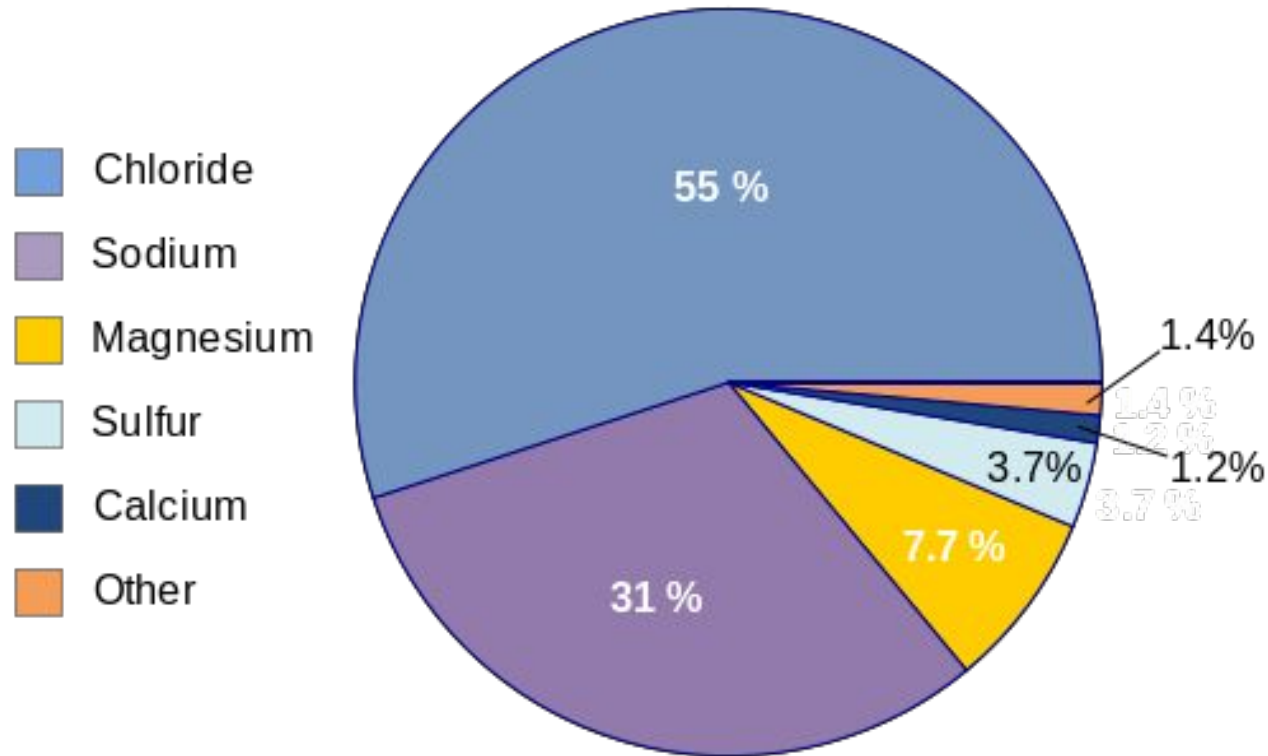


Light

Pressure



Seawater consists of water and other compounds



Salty water is denser than freshwater

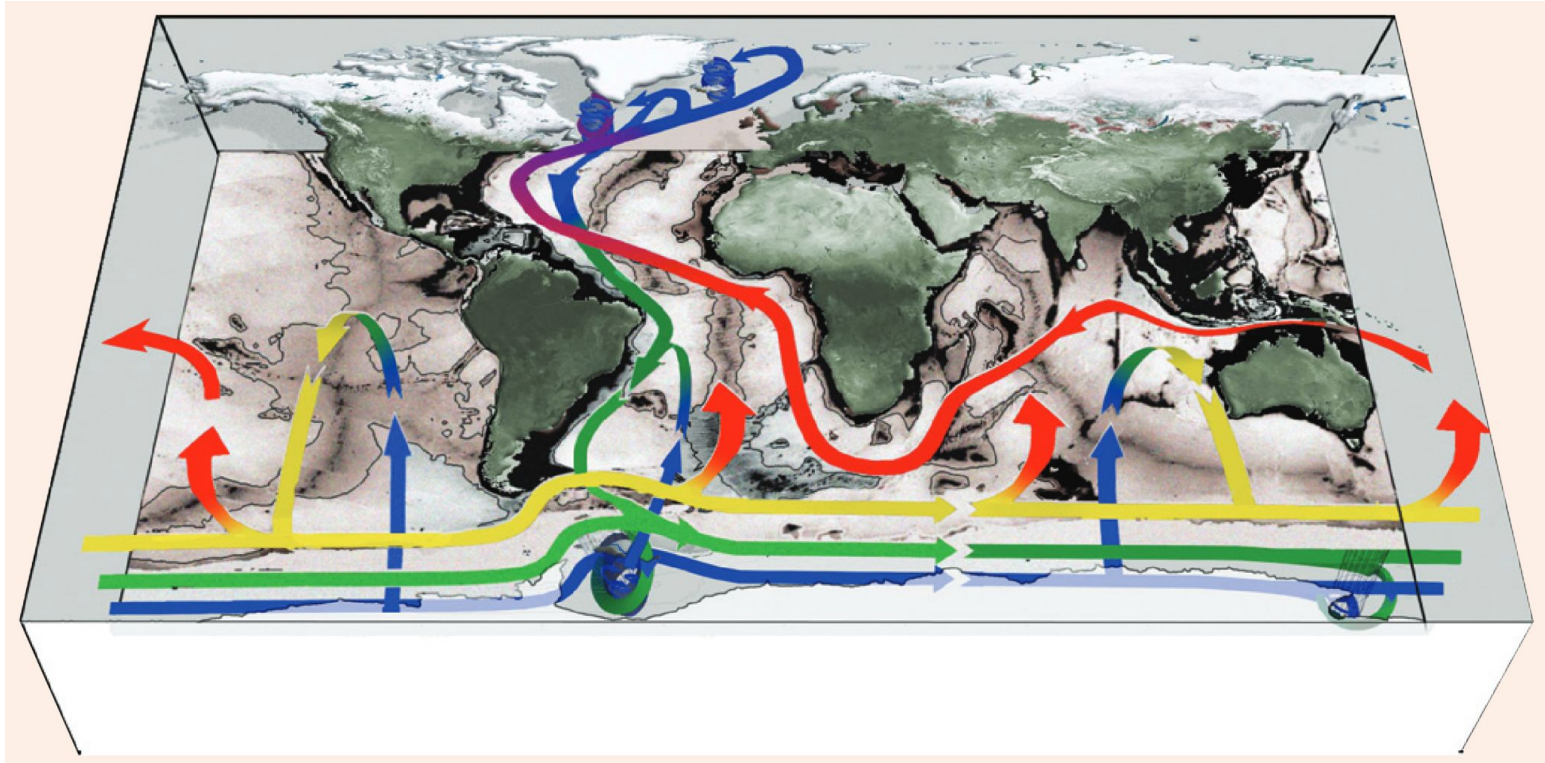


Fresh water



Salt water

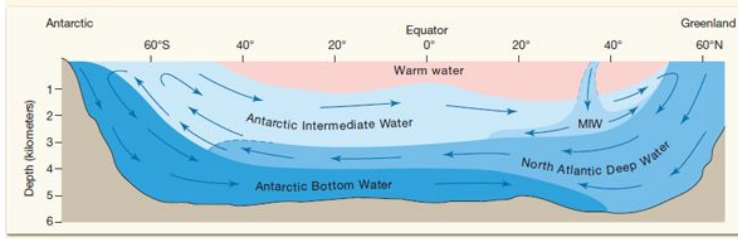
The ocean circulation is an emergent global phenomenon



[Marshall & Speer, 2012]

Cold Hot

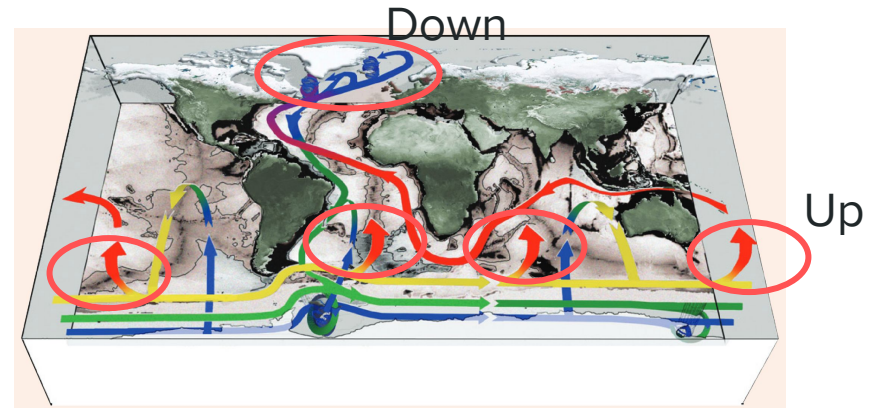
A global “conveyor”



- Asymmetrical heating/cooling, along with brine rejection from ice formation create the Thermohaline circulation
- Referred to as the ‘meridional overturning circulation’

Changes in overturning circulation have **big** climate impact

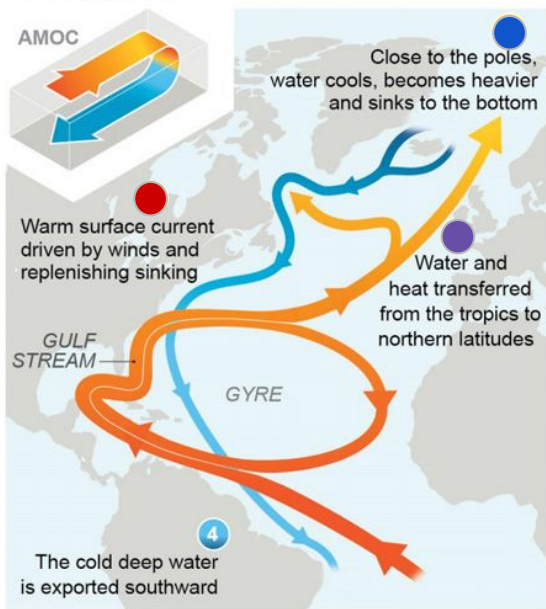
- The ocean is layered with the heaviest/densest water at the bottom
- Intense cooling and ice formation at the poles create very dense waters



Major currents may change in a warmer climate: Gulf Stream example

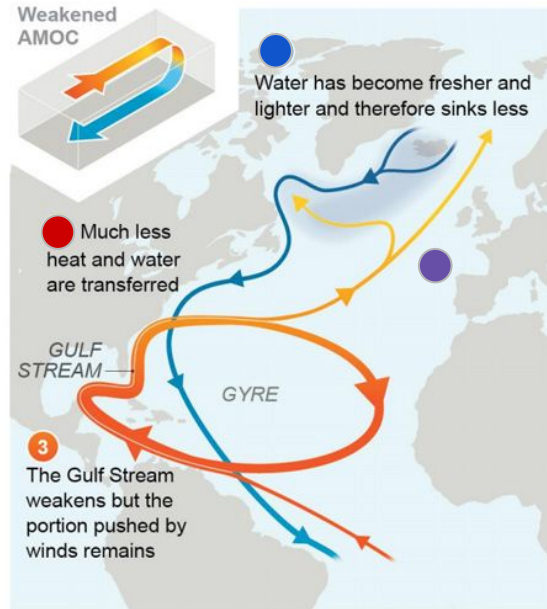
Today

The Gulf Stream is part of both the horizontal, subtropical gyre and the vertical, Atlantic Meridional Overturning Circulation (AMOC)



In a warmer world

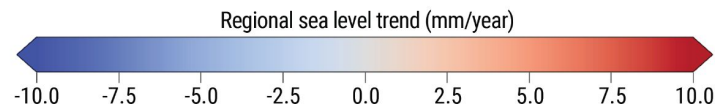
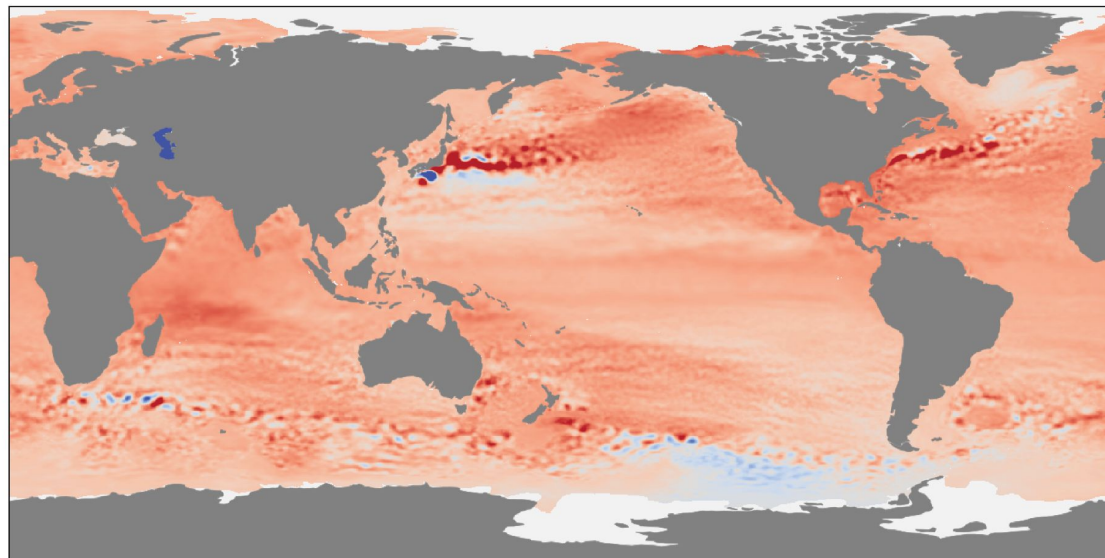
Climate change weakens the AMOC, which slows the Gulf Stream down



- Gulf stream shifts north
- Path of heat delivery moves
- Less dense/heavy waters

Sea level trends from February 1999 to May 2025

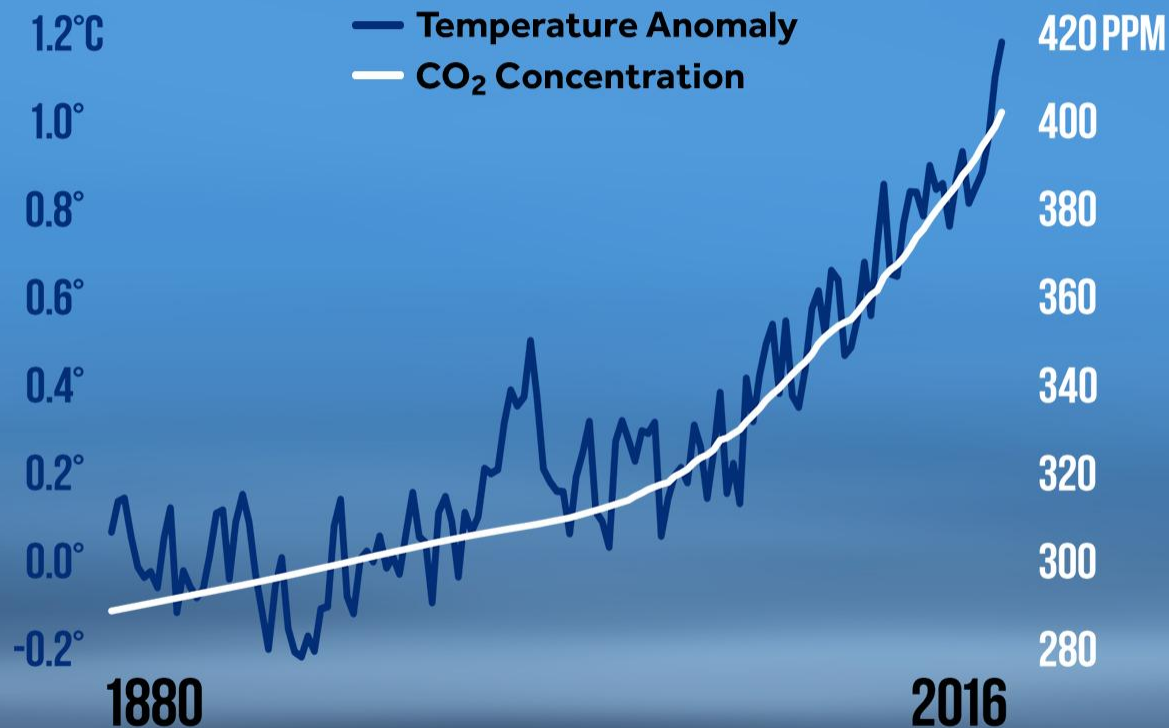
Data: CMEMS Ocean Monitoring Indicator based on C3S sea level product • Credit: C3S/ECMWF/CMEMS



Sea level rise has several sources



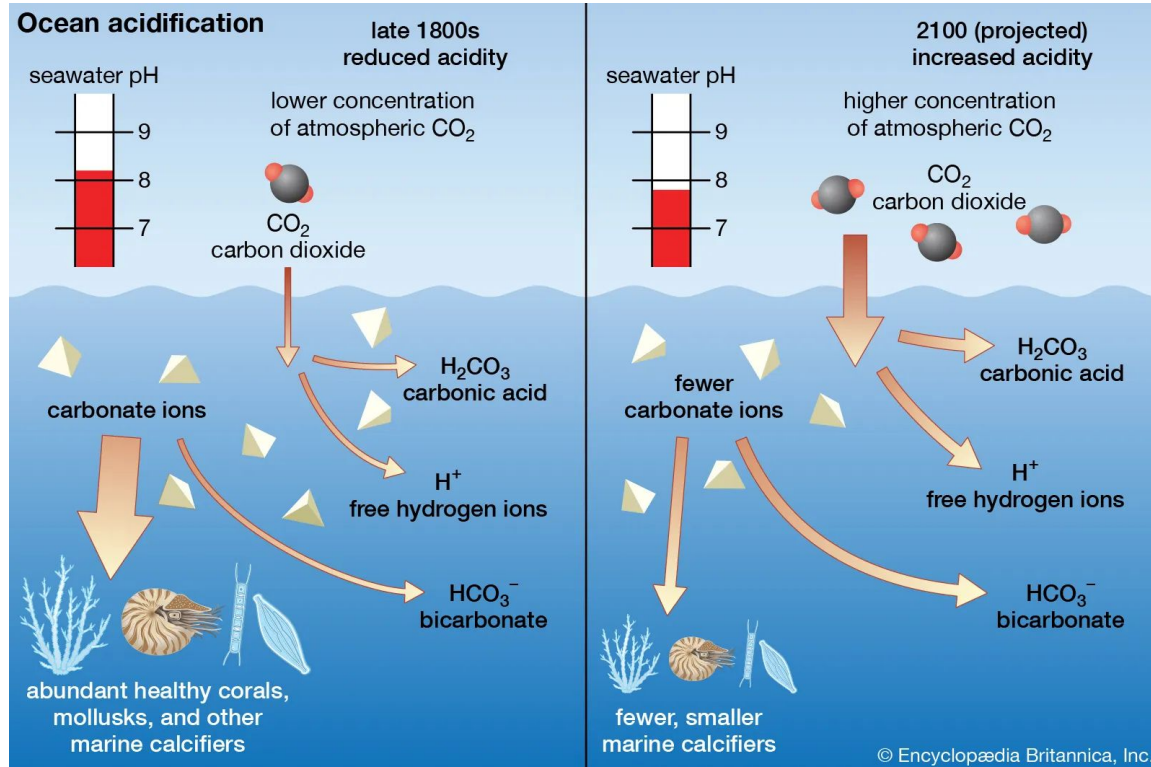
Global Temperature and Carbon Dioxide



Global temperature data averaged and adjusted to early industrial baseline (1881-1910).
Source: NASA GISS, NOAA NCEI, ESRL

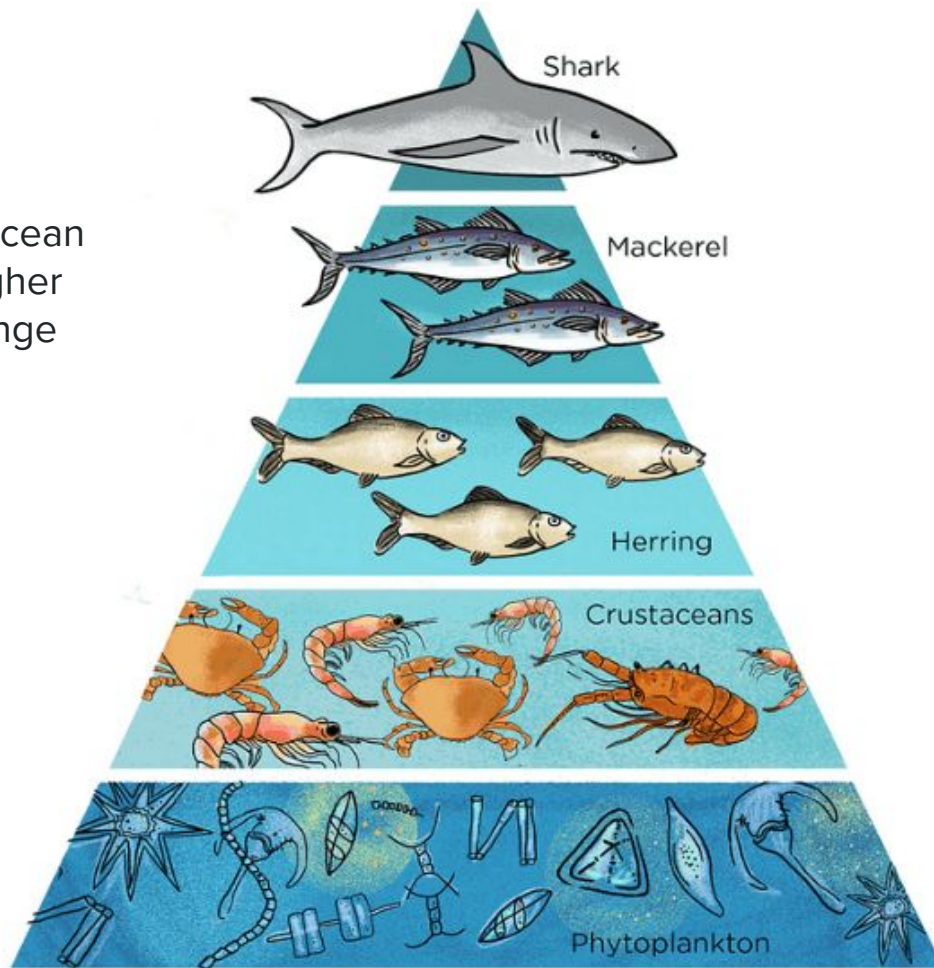
CLIMATE  CENTRAL

Climate change will make the ocean more acidic



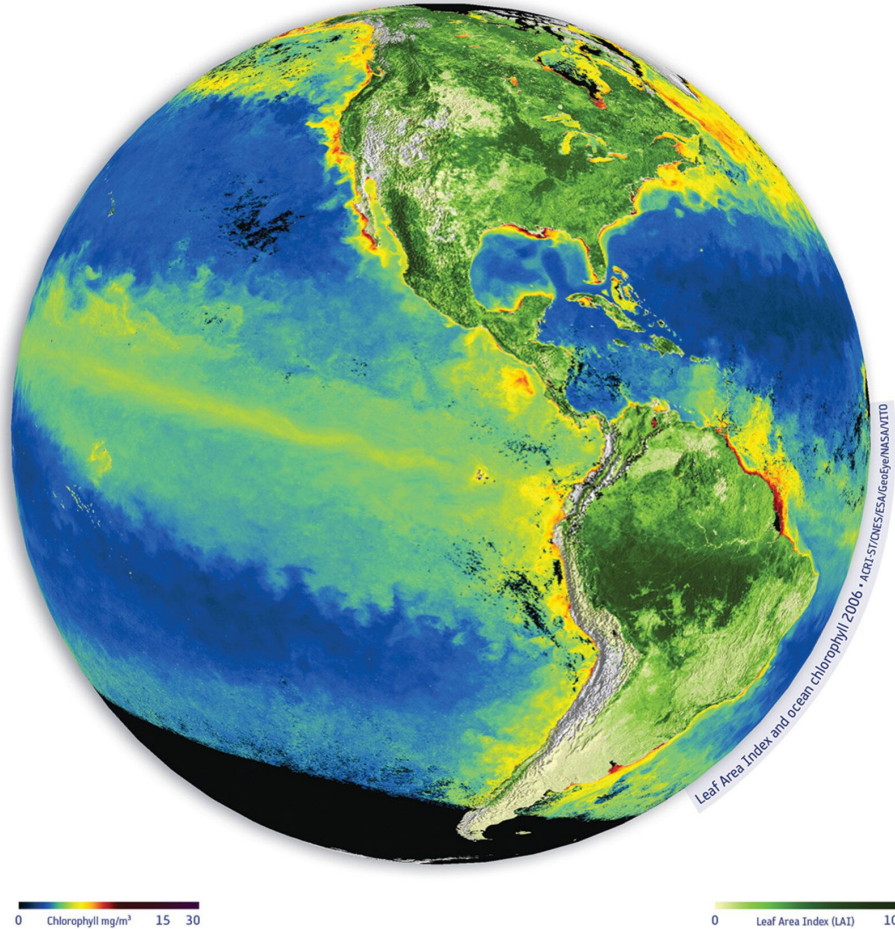
Marine life

The food-web rests on ocean “plants”, with species higher up comprising a wide range



The base: Plankton

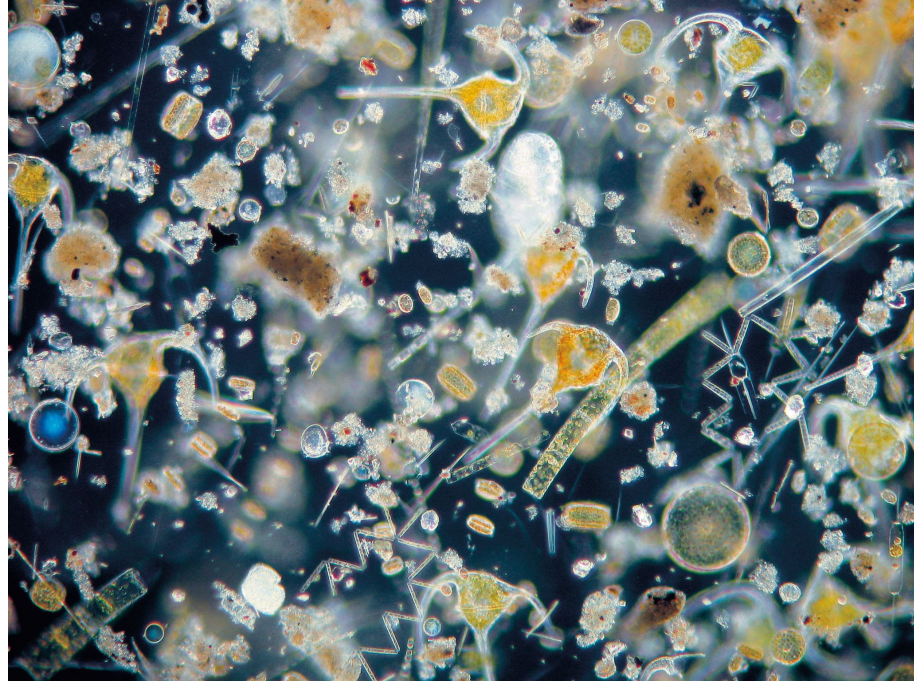
Just like terrestrial plants, plankton produce chlorophyll and rely on sunlight and nutrients to grow and there are many types



The base: Plankton

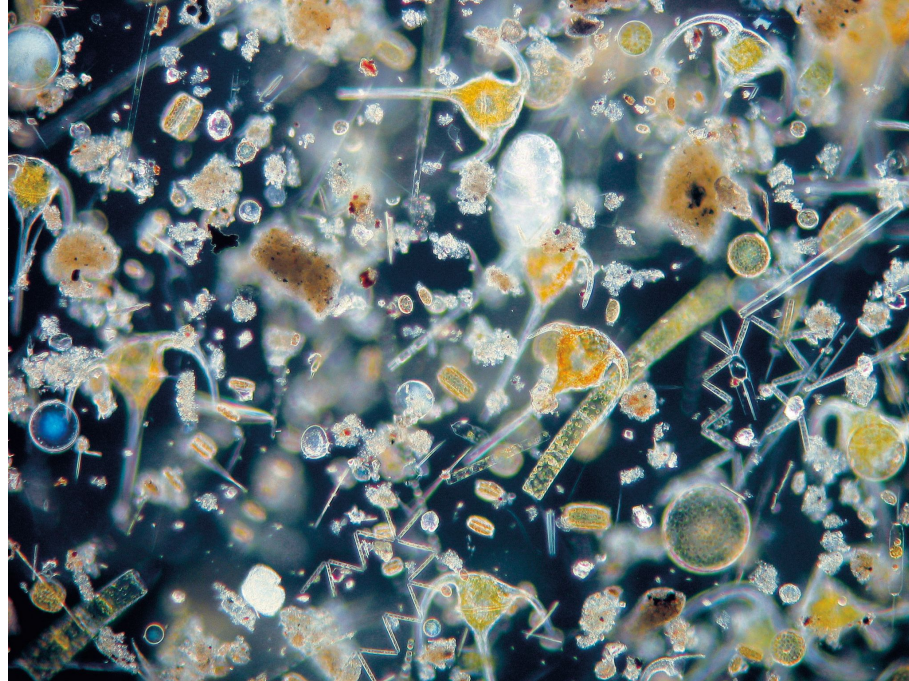
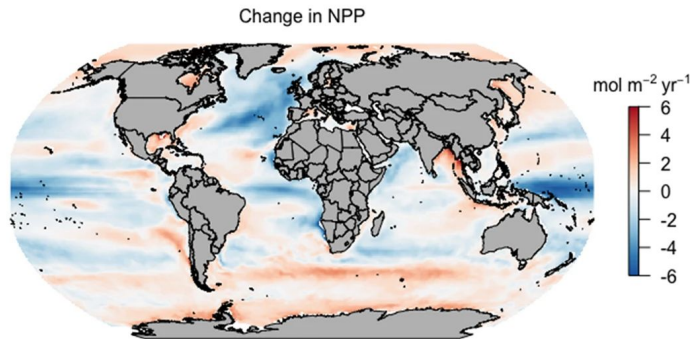
The plethora of types range orders of magnitude in size and have different “niches” expressed, for example, in what they need to flourish:

Some build shells, and some make their own nutrients!



The base: Plankton

Net primary production (NPP) is predicted to change drastically



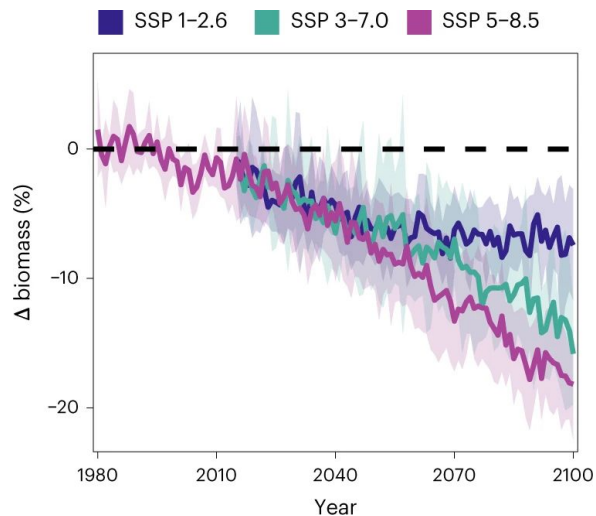
The base: Zooplankton

Microscopic organisms that “graze” on other plankton also range widely in size and form



Fisheries and conservation

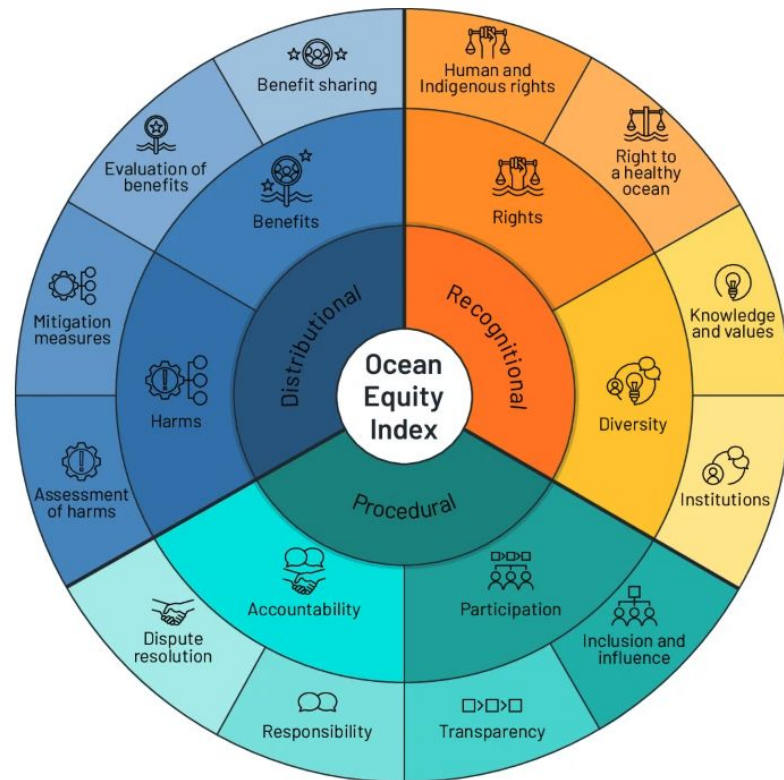
Fish rely on zooplankton for food



Equity and stakeholders

Who carries risk? Who benefits?

Some communities are more vulnerable than others, for example, living in low-lying coastal states, or in fisheries-dependent livelihoods, and may not have finances for adaption

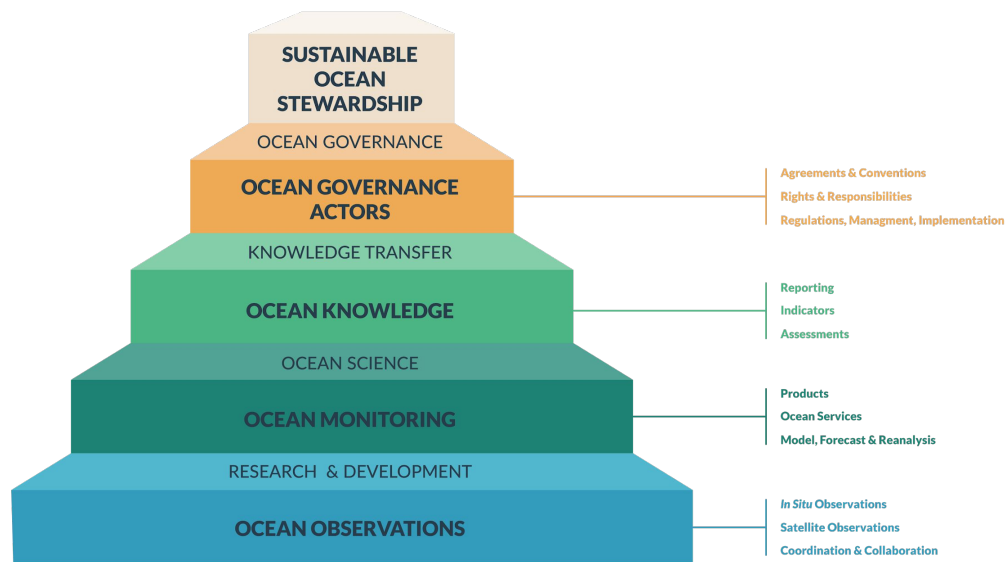


Ocean resources and governance

Ocean resources span a wide range of activities including fishing, tourism, shipping, and environmental protection

Most governments have individual policies, together with some regional and global governing bodies, and laws governing international waters

Global initiatives exist for ocean conservation, such as the 30 by 30 to designate 30% of Earth's land and ocean area as protected areas by 2030

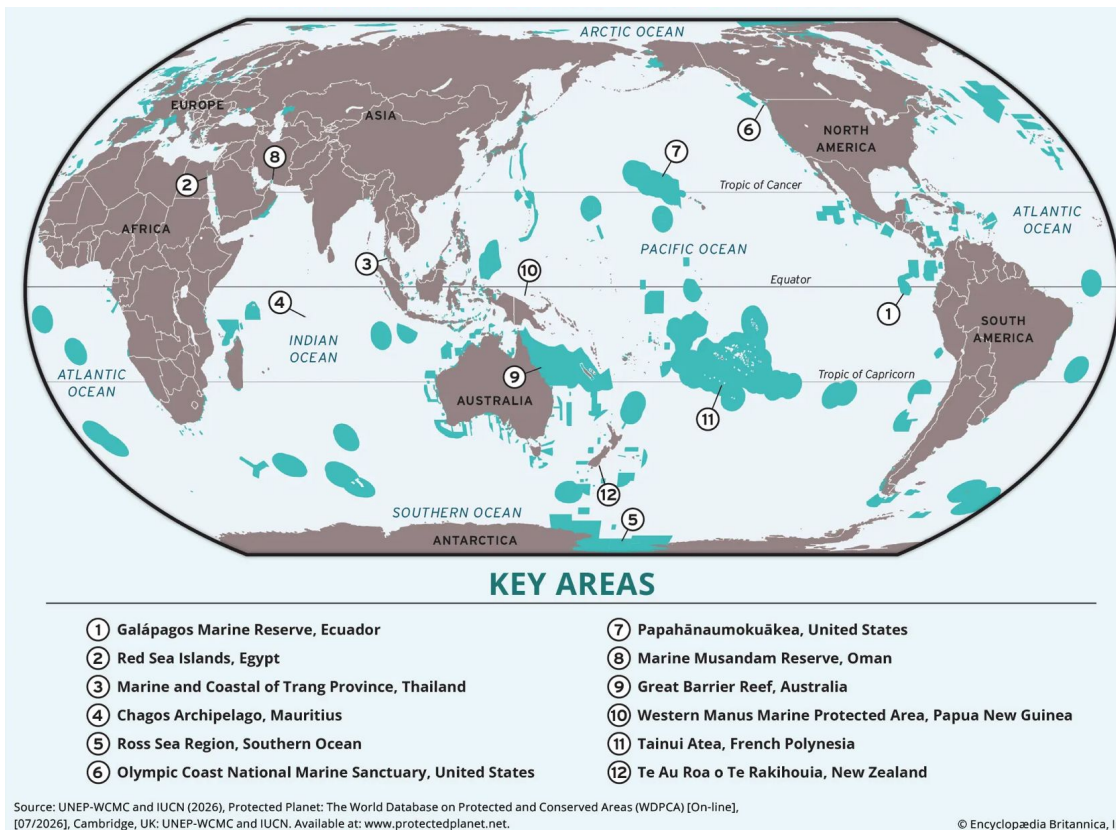


2. How much of the ocean do you think is currently a Marine Protected Area able to protect biodiversity?



<https://partici.fi/86456312>

Marine Protected Areas

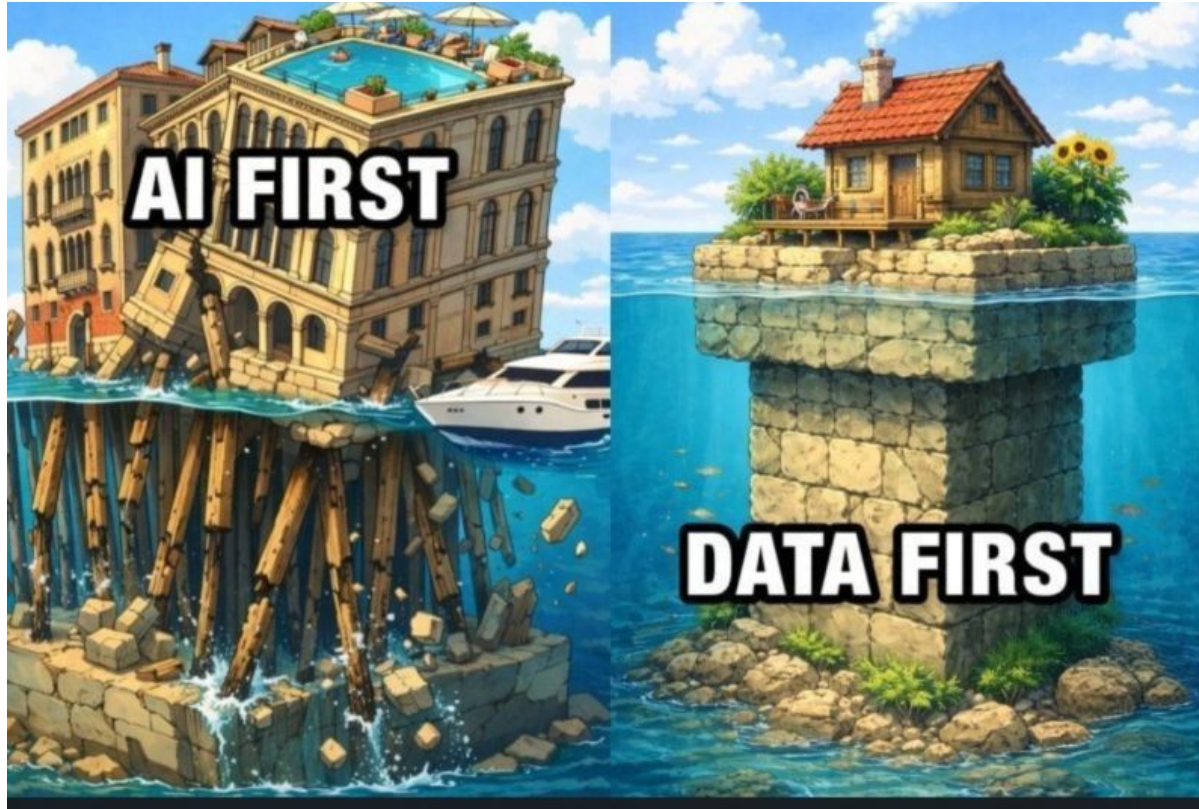


A data revolution, or a data problem?

Are we data rich and insight
poor, or data poor and rich in
insight?



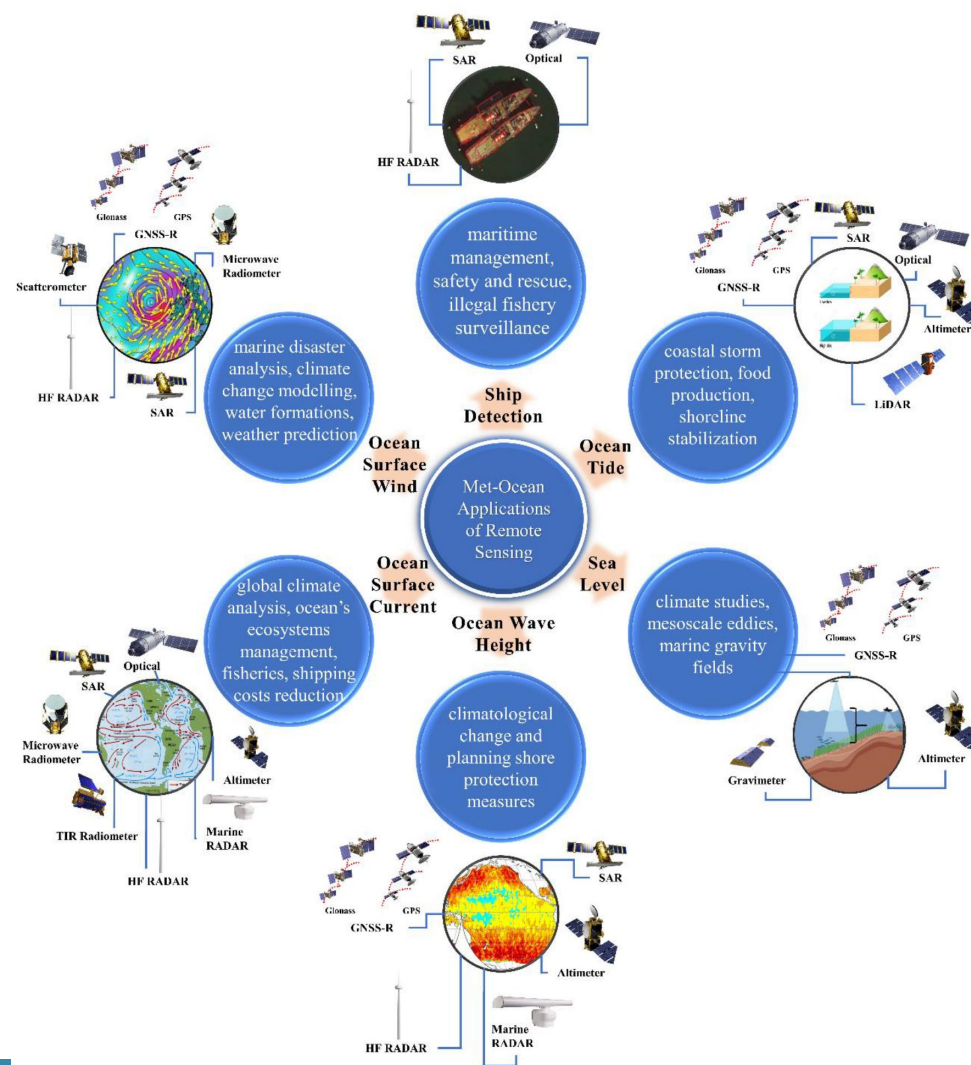
Our AI is only as powerful as our data



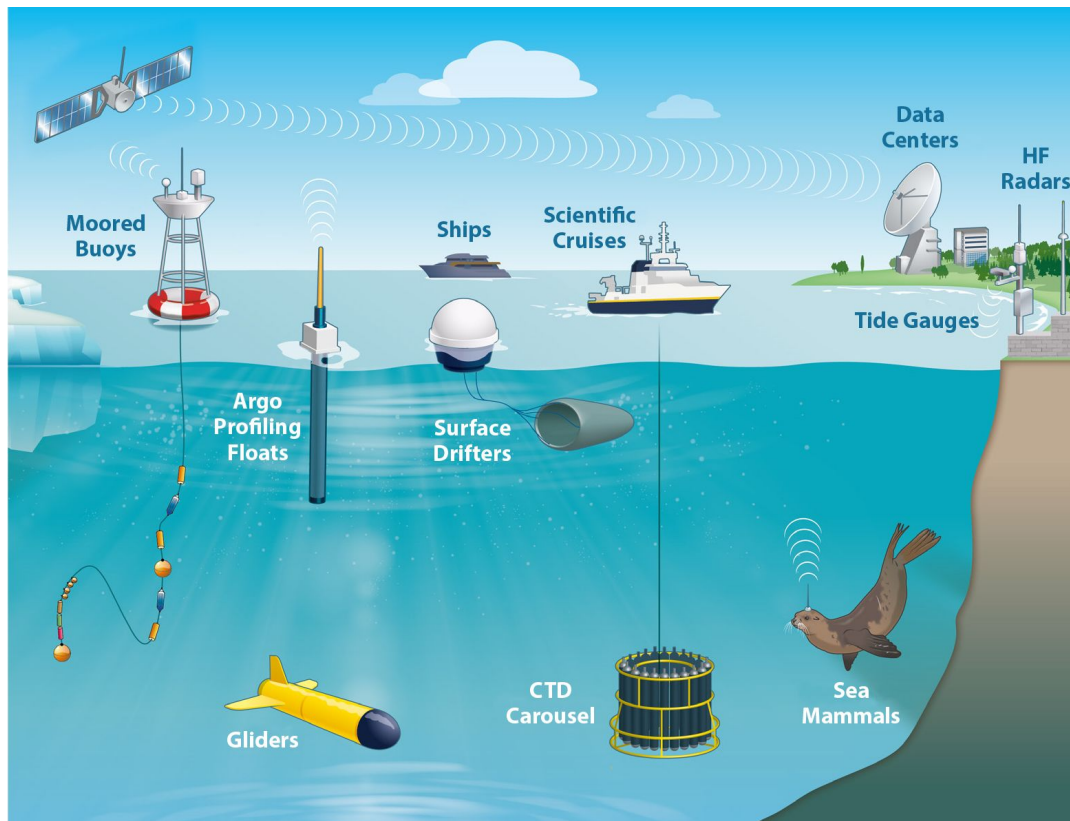
Observations: Collected “in-situ”



Observations: Remote sensing



Observations: A global network



Observations can come with a little help...

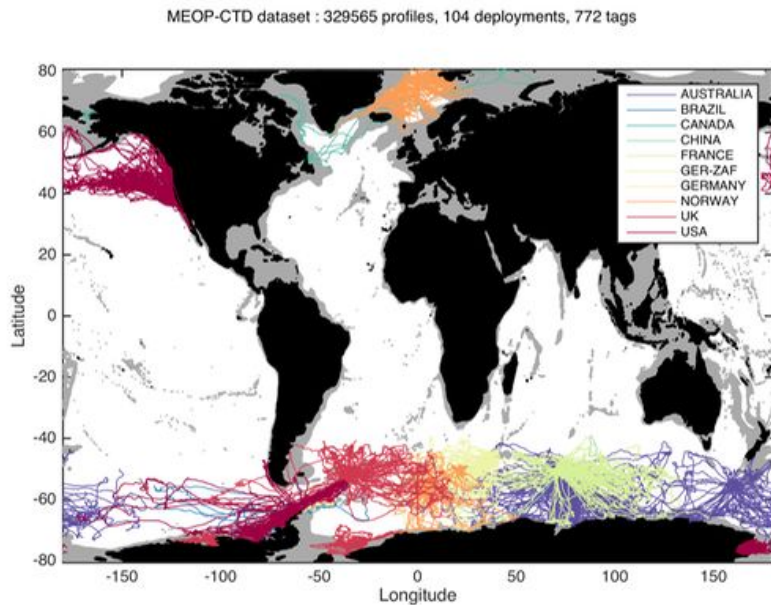
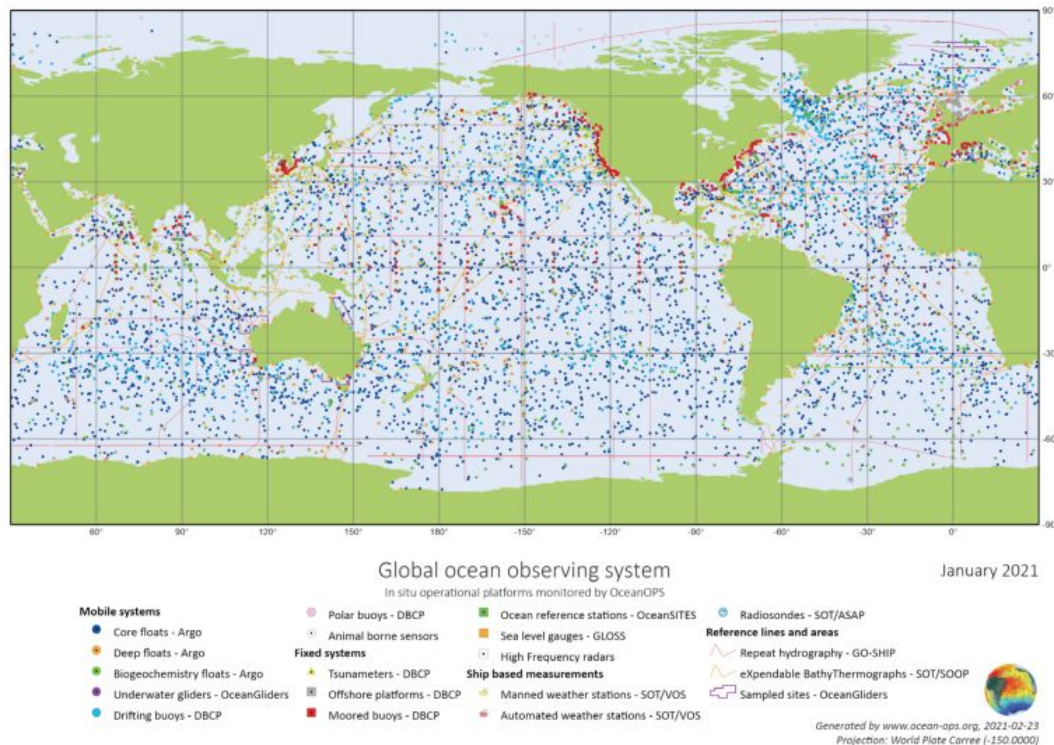
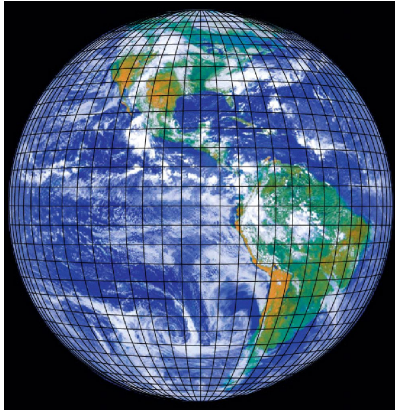


Photo: D. Costa

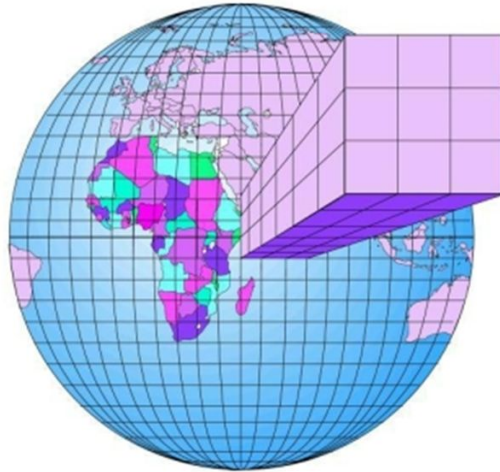
Equity in observations?



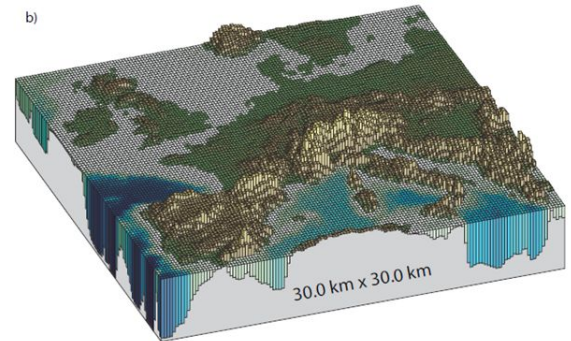
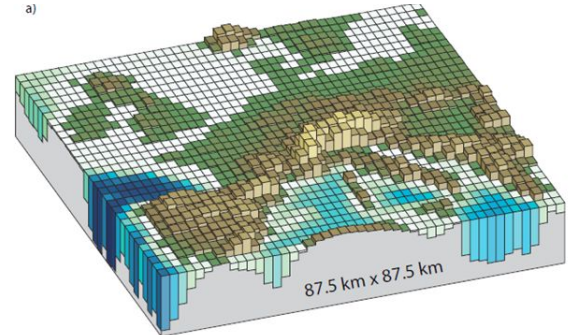
Numerical models



Grids extend downward into the land surface and oceans.



Just like on television, higher resolution means more realism



Numerical models: Primitive equations

Navier-Stokes + equation of state: Thin rotating fluid on a sphere

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} - f v = -\frac{1}{\rho_o} \frac{\partial P}{\partial x} + \frac{\partial}{\partial z} \left(K_v \frac{\partial u}{\partial z} \right) + F_u, \quad \text{Momentum (u component)}$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + f u = -\frac{1}{\rho_o} \frac{\partial P}{\partial y} + \frac{\partial}{\partial z} \left(K_v \frac{\partial v}{\partial z} \right) + F_v, \quad \text{Momentum (v component)}$$

$$\frac{\partial P}{\partial z} = -\rho g, \quad \text{density}$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0, \quad \text{continuity}$$

$$\frac{\partial \theta}{\partial t} + u \frac{\partial \theta}{\partial x} + v \frac{\partial \theta}{\partial y} + w \frac{\partial \theta}{\partial z} = \frac{\partial}{\partial z} \left(K_h \frac{\partial \theta}{\partial z} \right) + F_\theta, \quad \text{Advection of temperature}$$

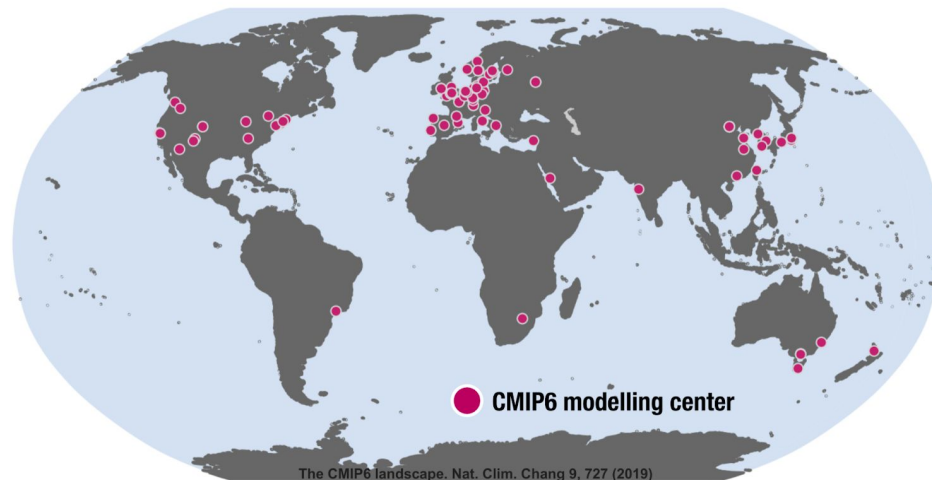
$$\frac{\partial s}{\partial t} + u \frac{\partial s}{\partial x} + v \frac{\partial s}{\partial y} + w \frac{\partial s}{\partial z} = \frac{\partial}{\partial z} \left(K_h \frac{\partial s}{\partial z} \right) + F_s, \quad \text{Advection of salinity}$$

$$\rho = \rho(\theta, s), \quad \text{Density as a function of } \theta \text{ and } s$$

The Climate Model Intercomparison Project



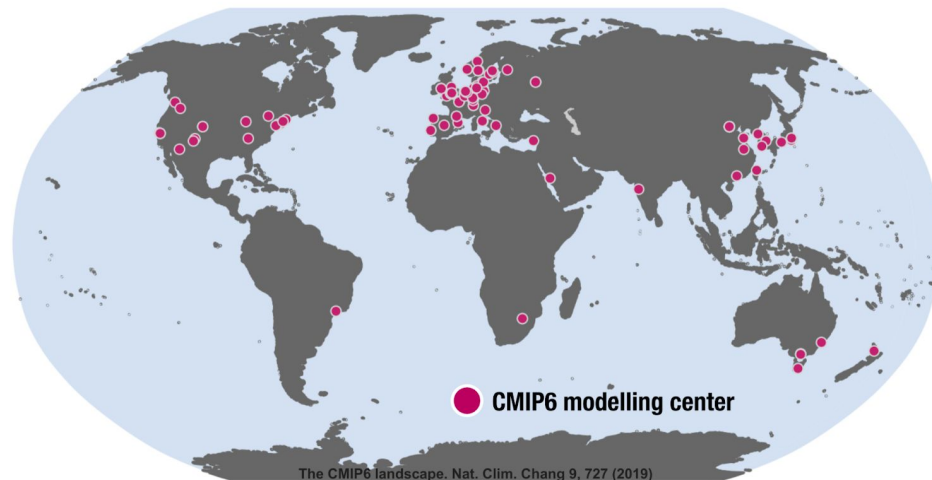
Modeling centers collaborate globally to run and share experiments of how societies' actions will impact climate



Equity in model development and data sharing?

Few models are developed in the Southern Hemisphere, so insight from these areas may be less well captured

Only a fraction of the data can actually be shared, so data to get insight on e.g. why the ocean is changing is difficult to get



Continued support for gathering data is vital

It is tempting to only focus on AI, but continuing support for improving the data AI relies on is crucial

Creating datasets with a balanced representation of locations and developers can improve equity



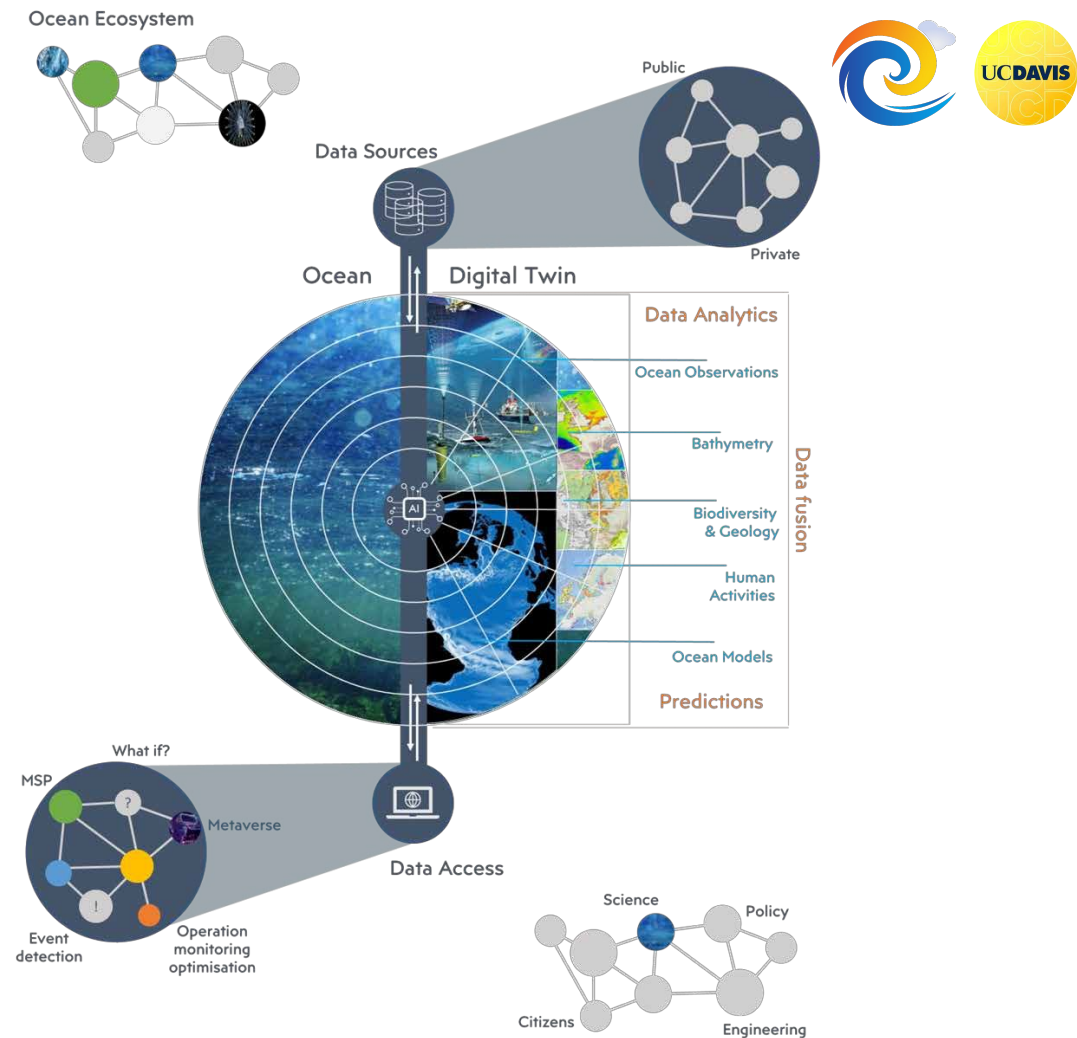
Break: What do you think most important to you to protect in the ocean?

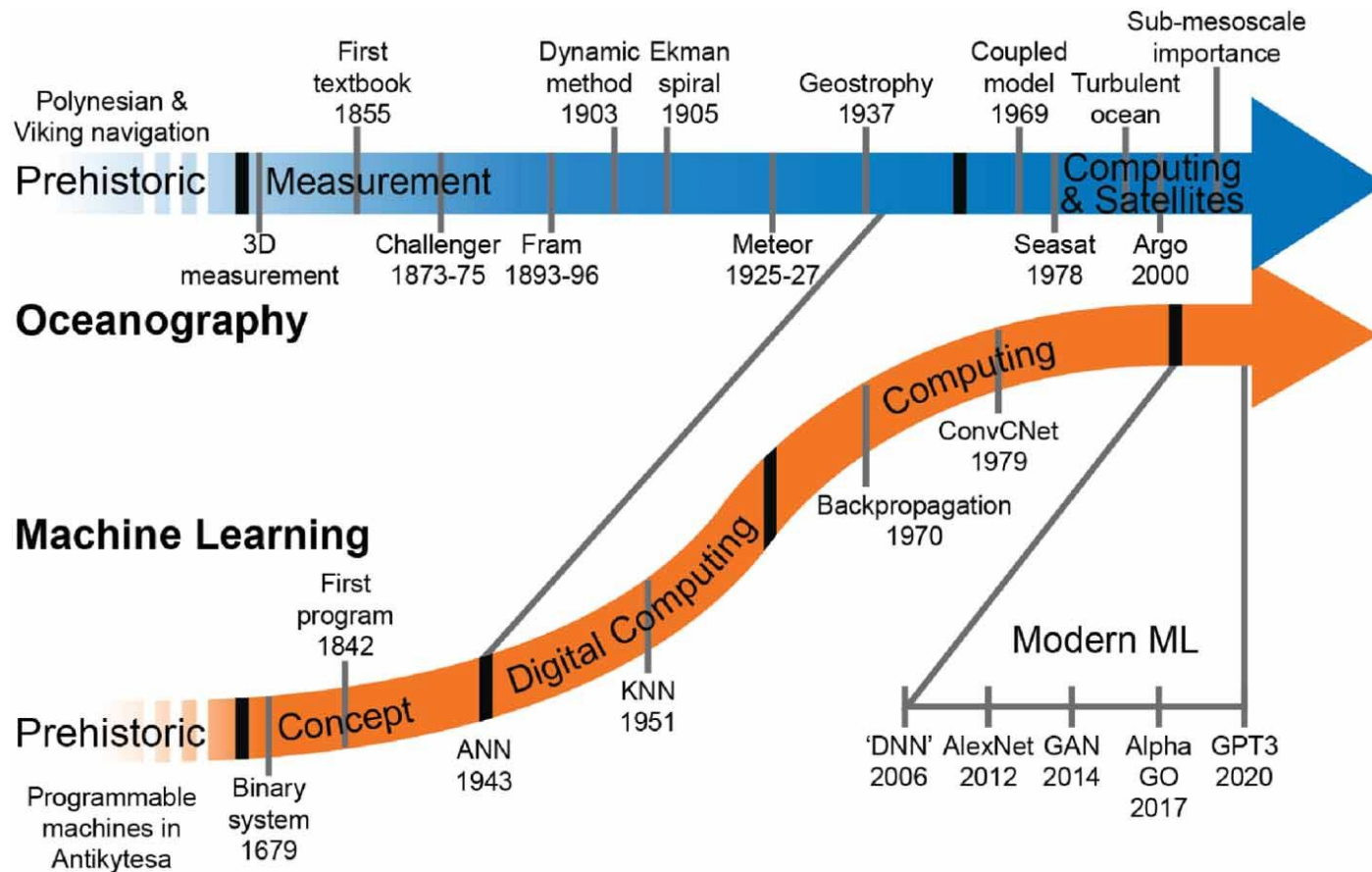


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AI for Oceans and Marine Science

How we can learn and help build resilience





ML: The missing piece?

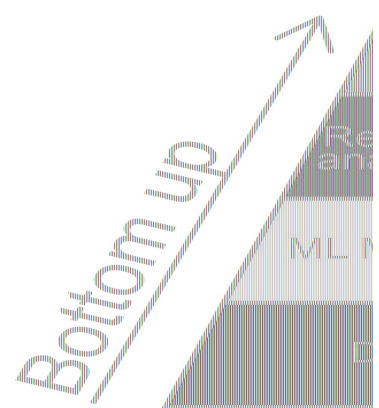
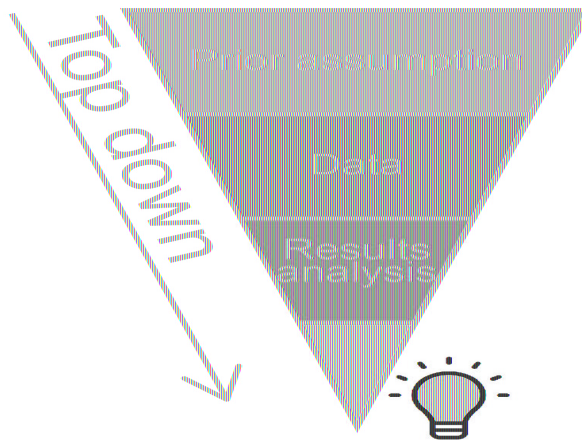
Too simple:

Theoretical descriptions tend to be oversimplified

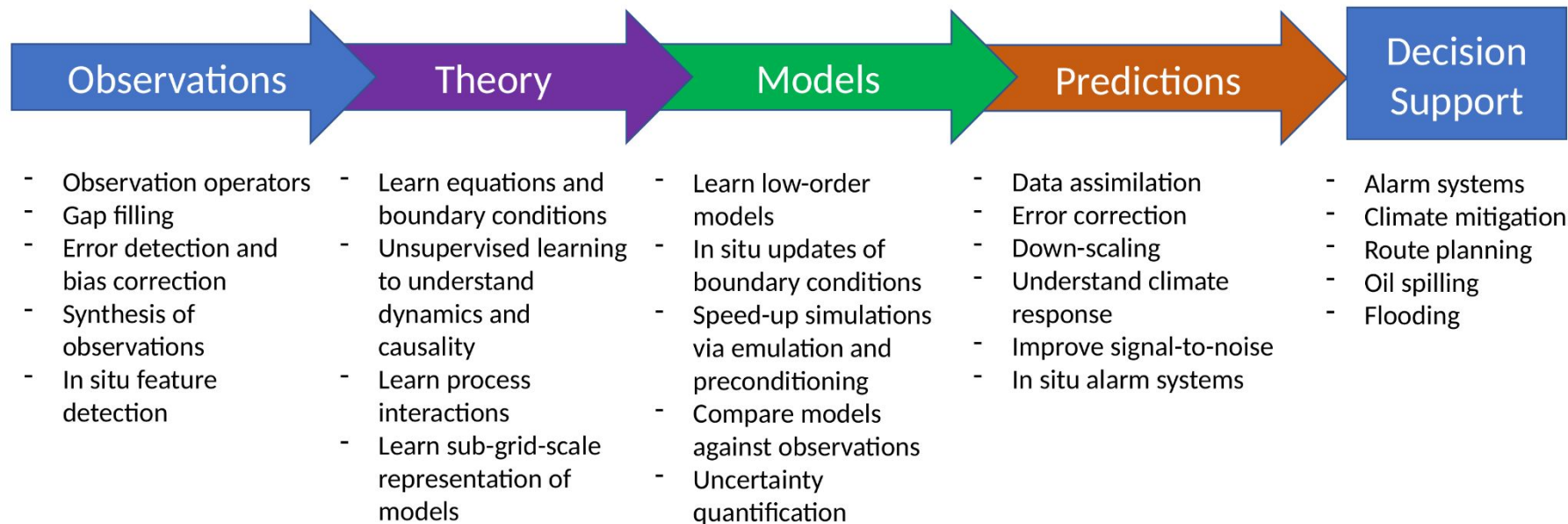
Too hard:

Inferring basic principles from data alone is extremely difficult

The bridge: ML

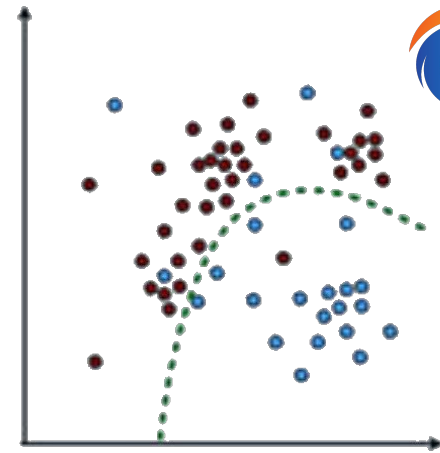
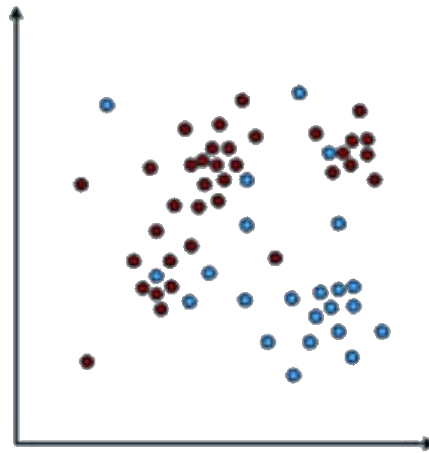


ML within oceanography



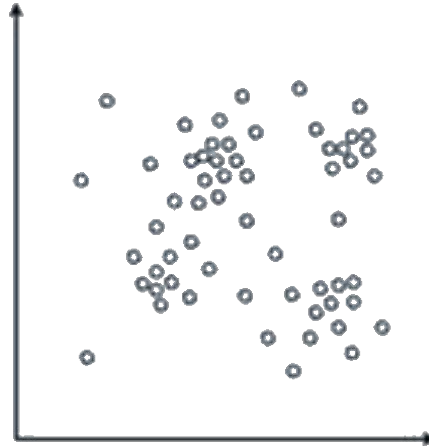
Supervised

- Labeled data
- Supervised unsupervised
- Decision boundary

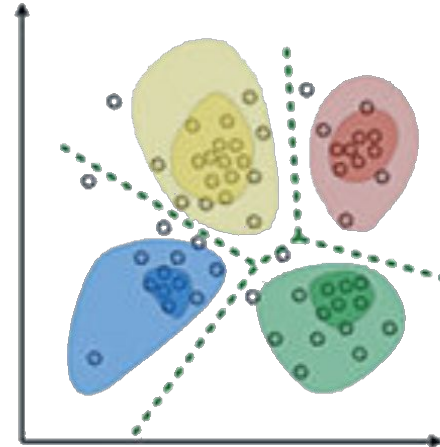


Unsupervised

- No labels
- Identify structures



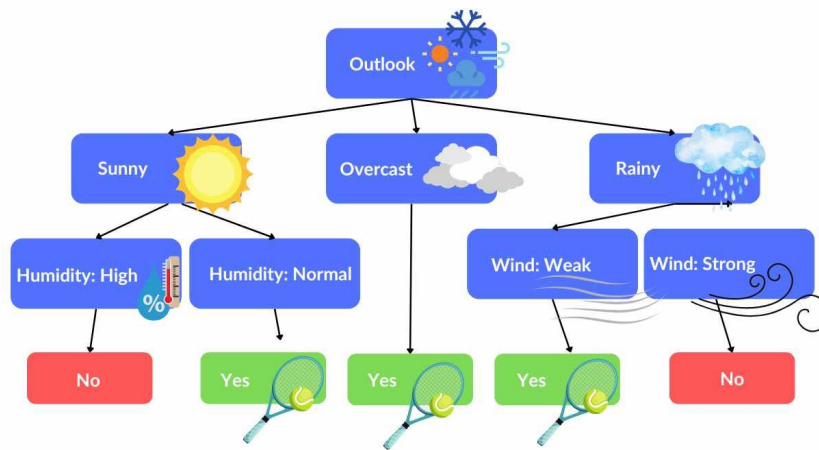
Training data



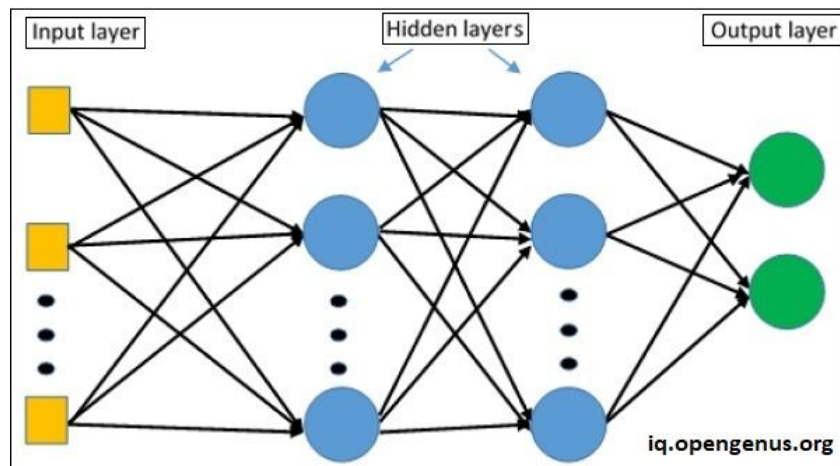
Resulting model

Two supervised ML methods

Random forests



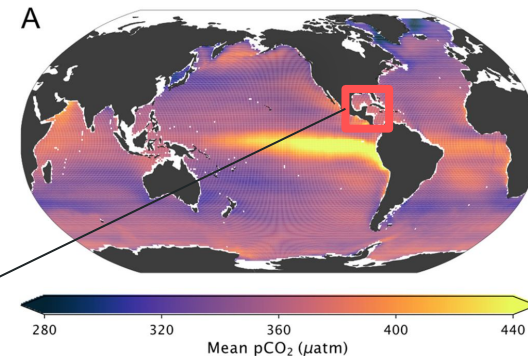
Neural networks



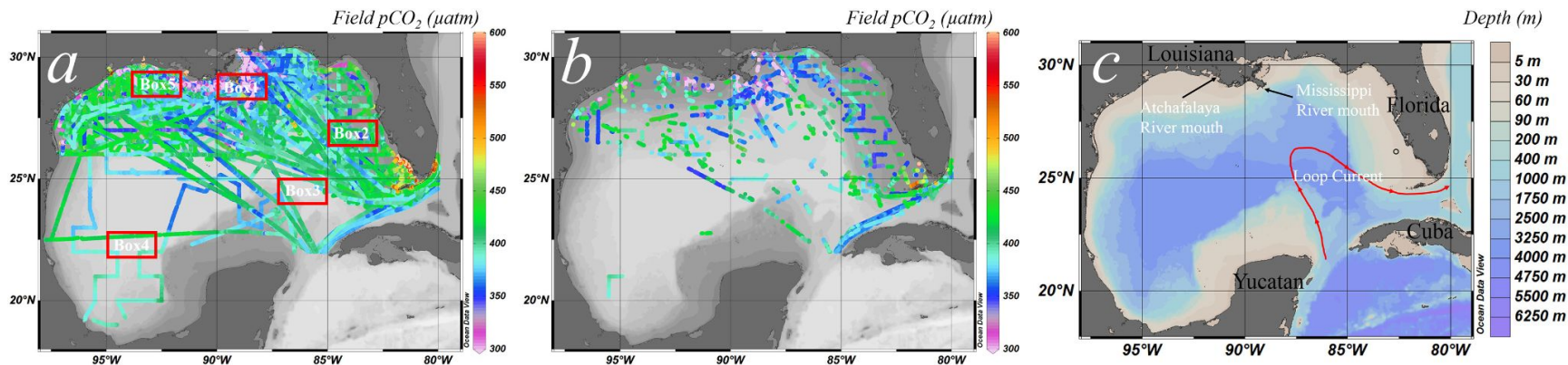
ML for monitoring ocean surface $p\text{CO}_2$

Measuring CO_2 requires great effort

There is no known function between another predictive variable and surface $p\text{CO}_2$



Gulf of Mexico

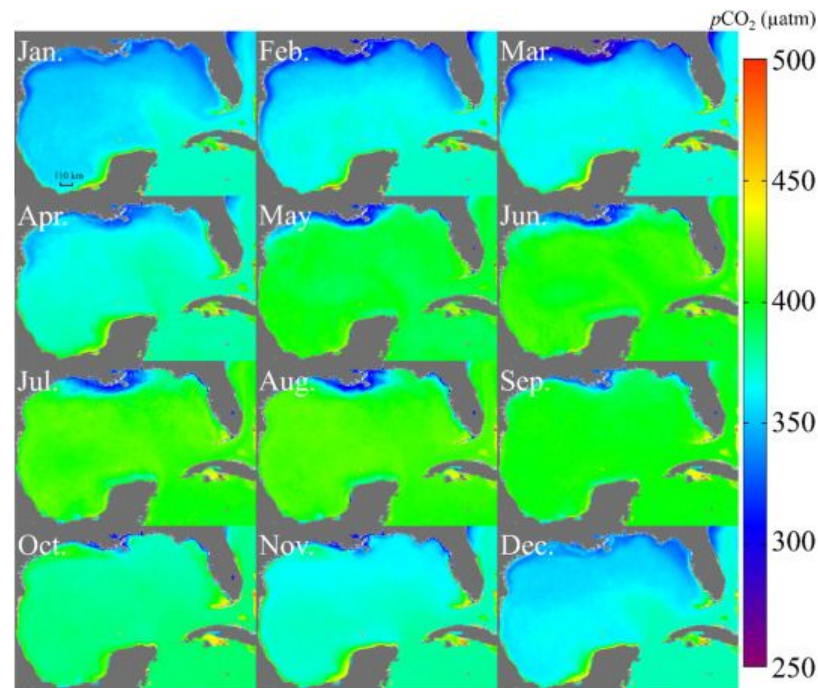


ML for monitoring ocean surface $p\text{CO}_2$

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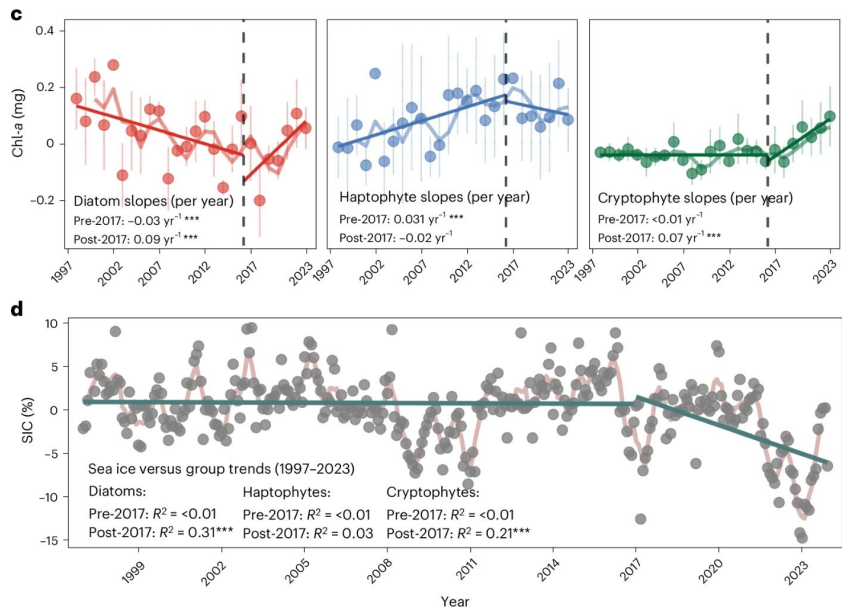
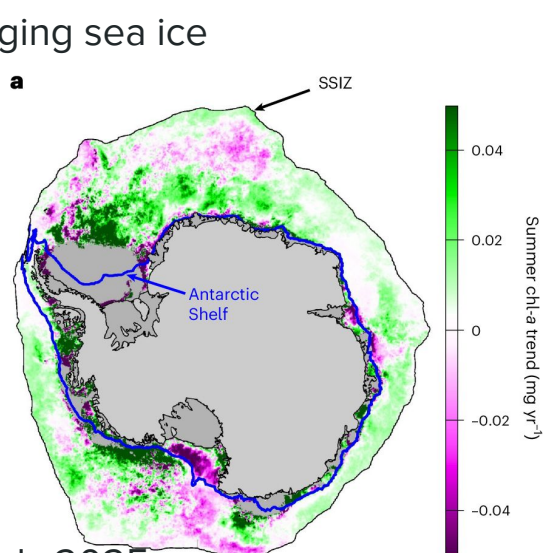
Chen et al., 2019 used an ML **random forest** method to have satellites predict $p\text{CO}_2$



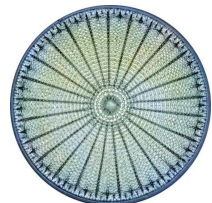
ML for monitoring Antarctic plankton

Phytoplankton are critical to the Antarctic marine food web and carbon sequestration, yet long-term shifts in their community composition are poorly understood

Hayward et al., 2025 used ML random forests to show how community composition has changed with changing sea ice



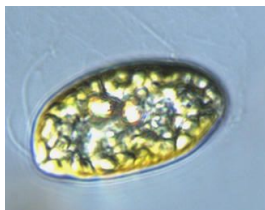
ML for monitoring Antarctic plankton



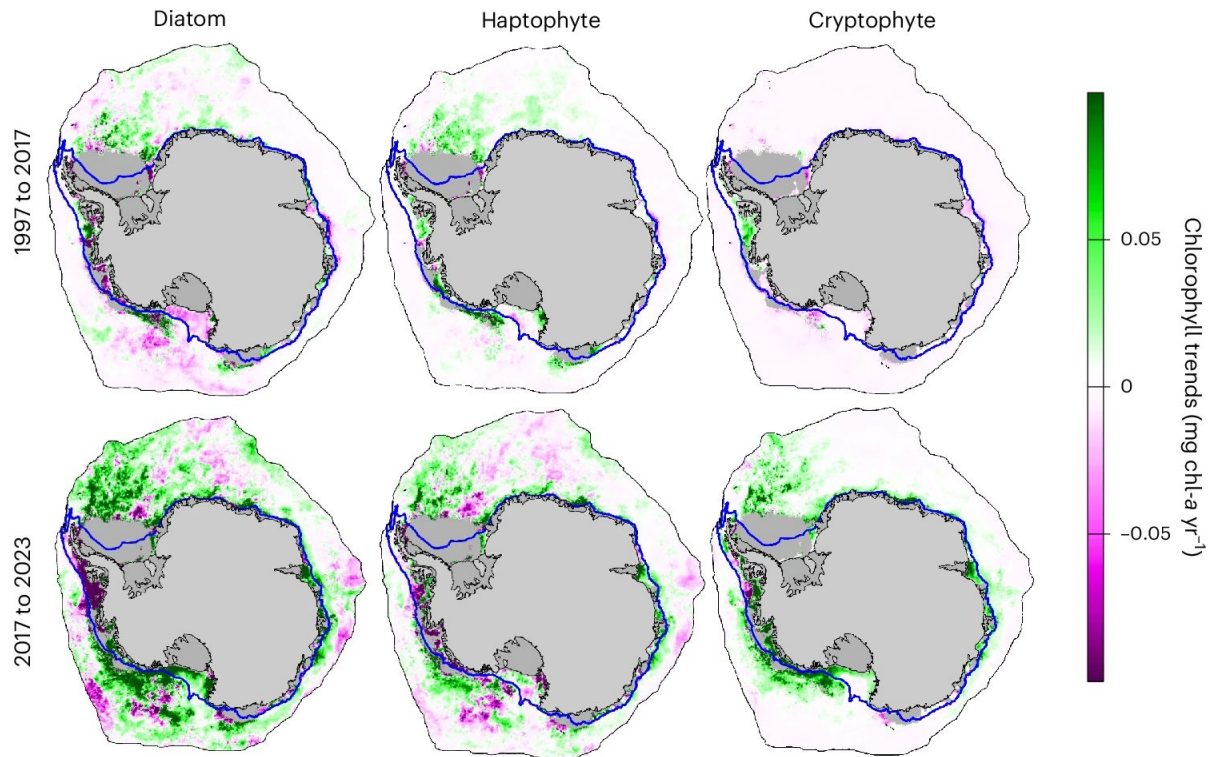
Diatoms



Haptophyte



Cryptophyte

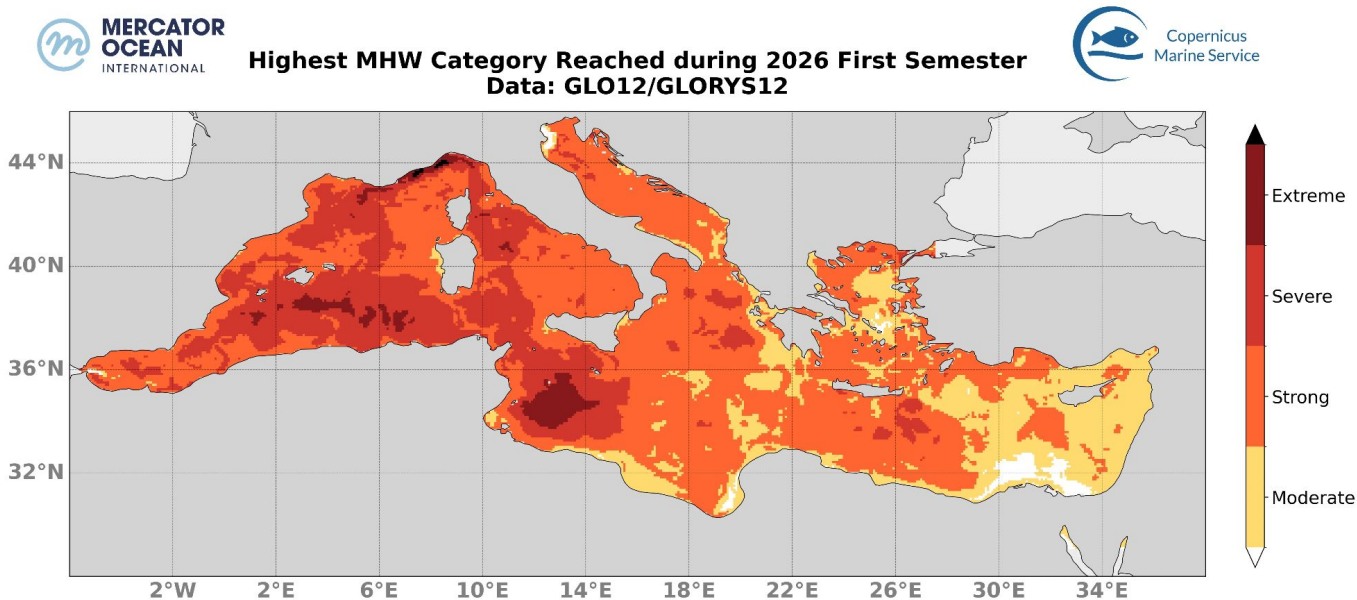


3: How much of the Mediterranean surface ocean experienced a heatwave during the first semester of 2026?

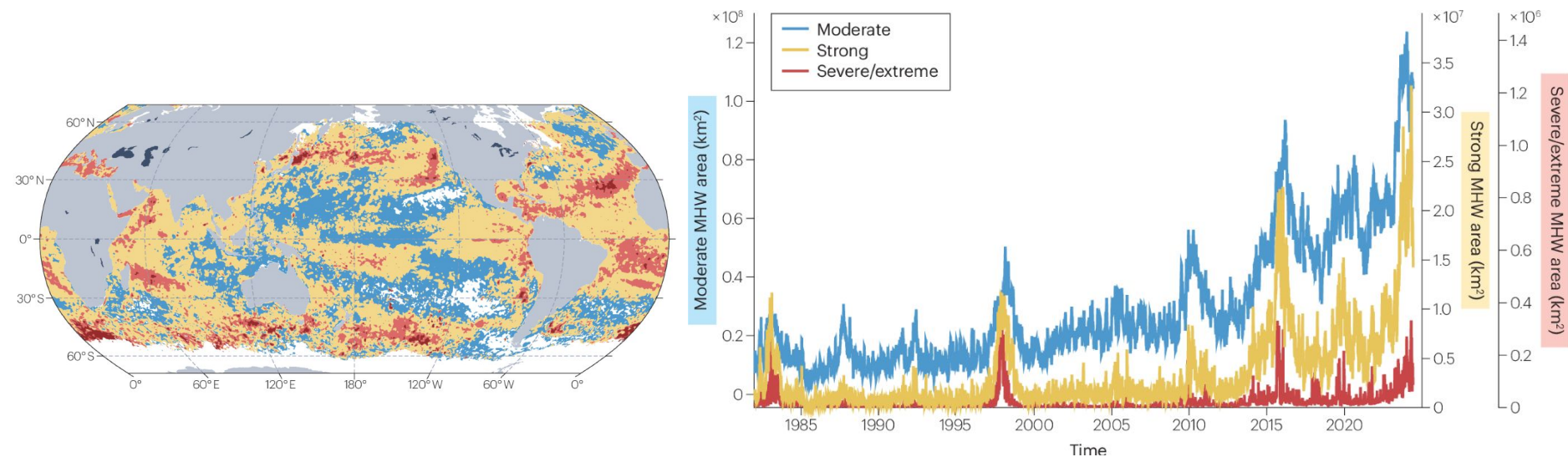


<https://partici.fi/86456312>

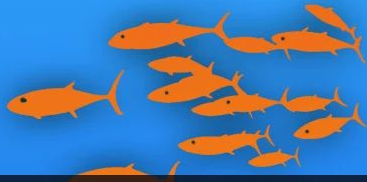
How much of the Mediterranean surface ocean experienced a heatwave during the first semester of 2026?



Marine heatwaves are becoming more common and severe



MARINE HEATWAVE IMPACTS



CHANGES RANGES
OF SPECIES' HABITATS



IMPACTS ECONOMICS
OF FISHERIES

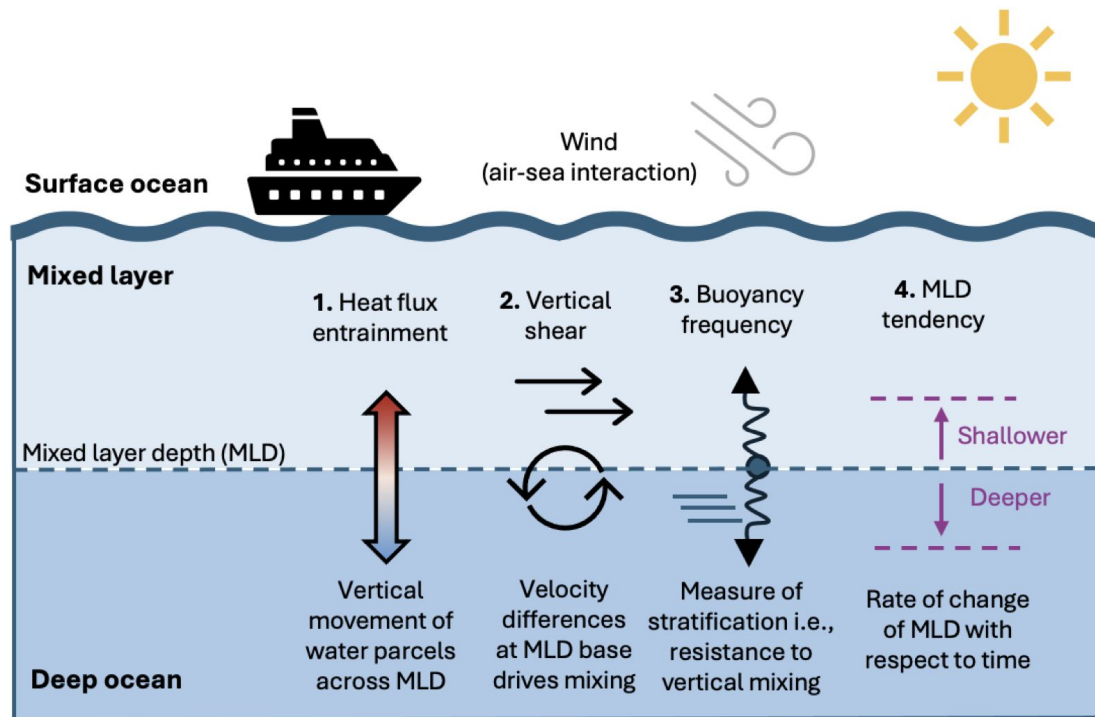


BOOSTS TROPICAL
STORMS & HURRICANES



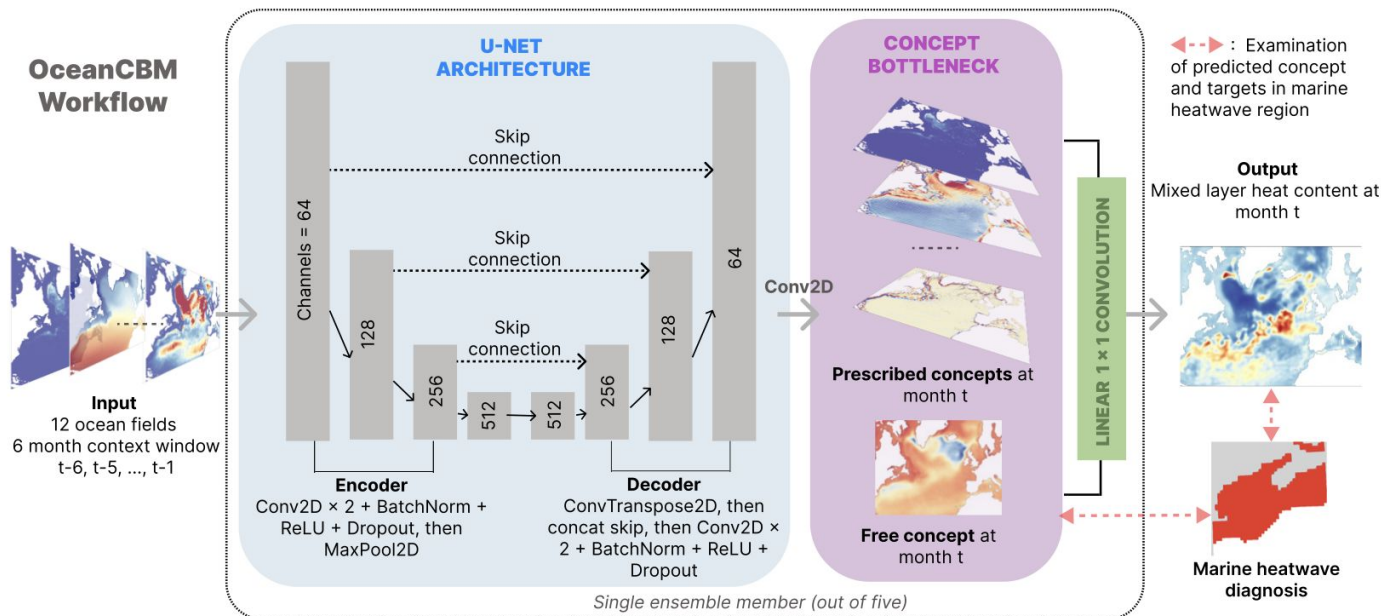
BLEACHES CORAL

ML for marine heatwaves prediction and attribution

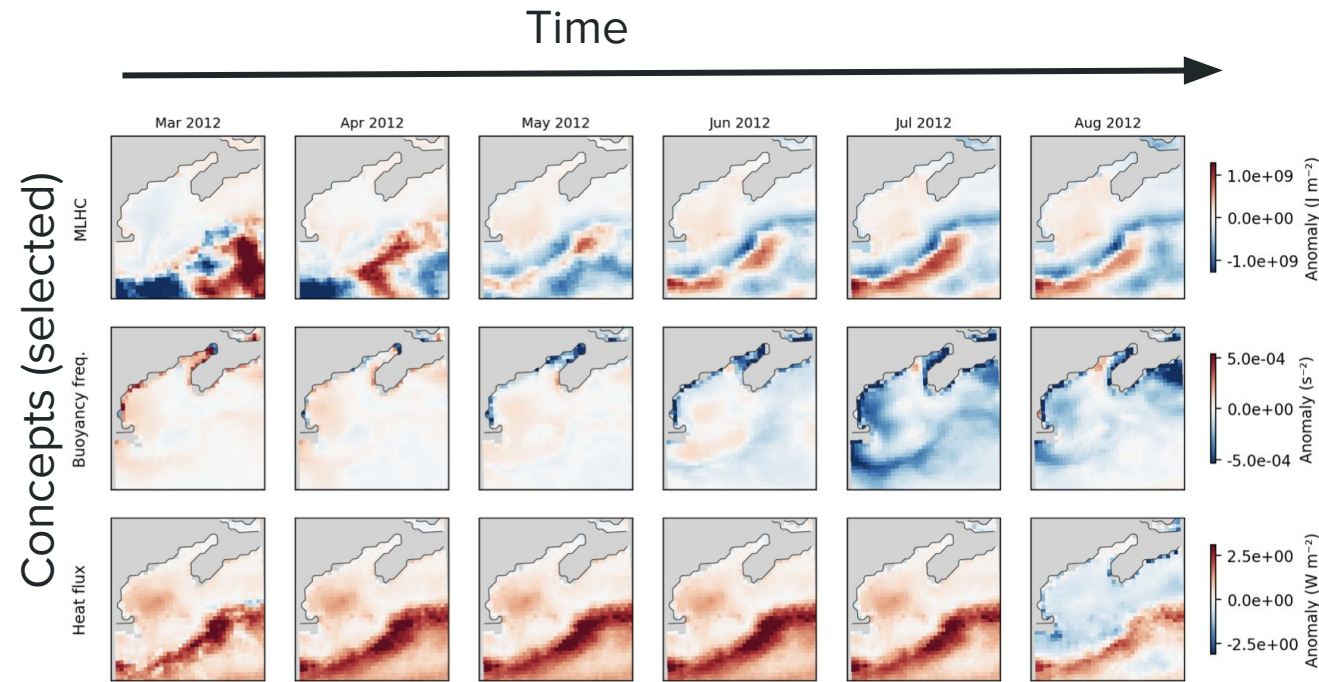


ML for marine heatwaves prediction and attribution

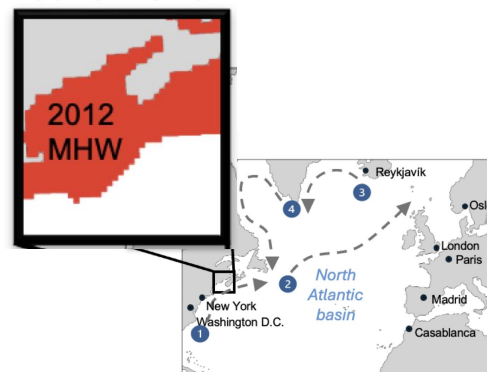
Predict ocean mixed layer heat content **telling network to choose** what was important



ML for marine heatwaves prediction and attribution



Gulf of Maine



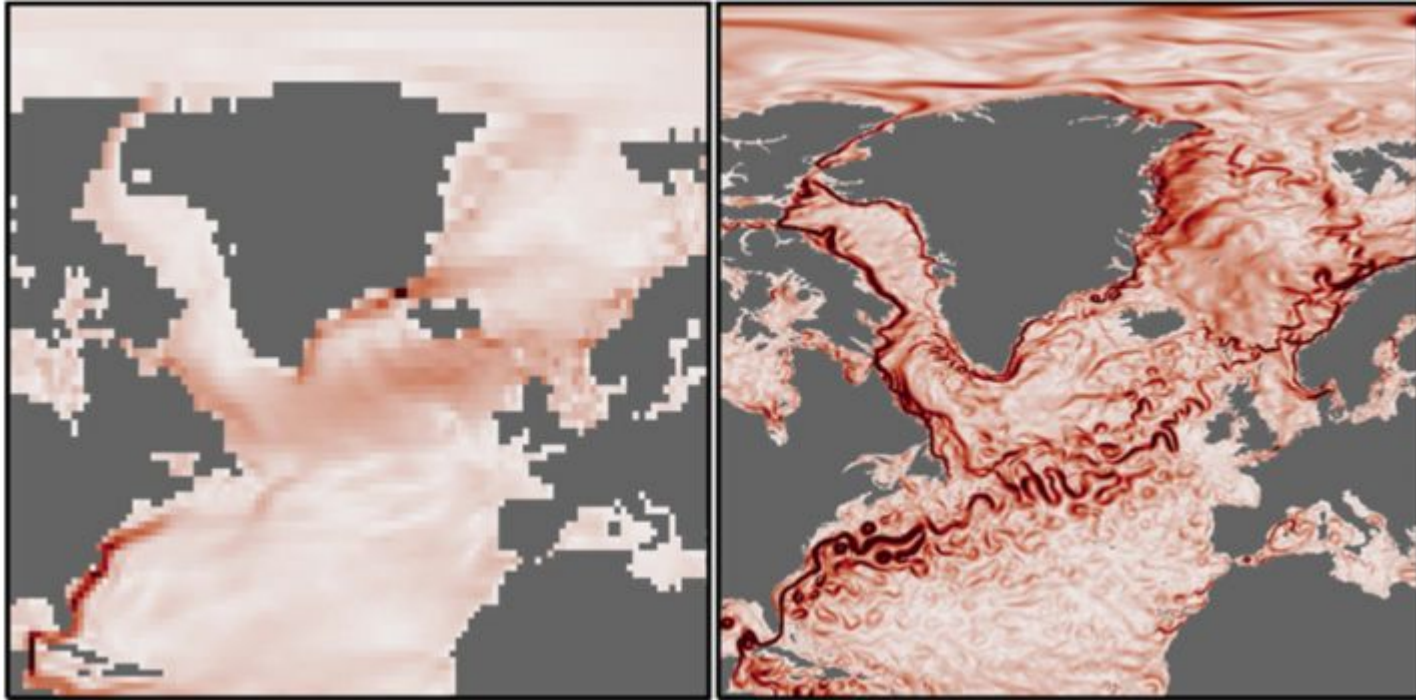
Conclusion:

Atmospheric heat input that warmed the mixed layer

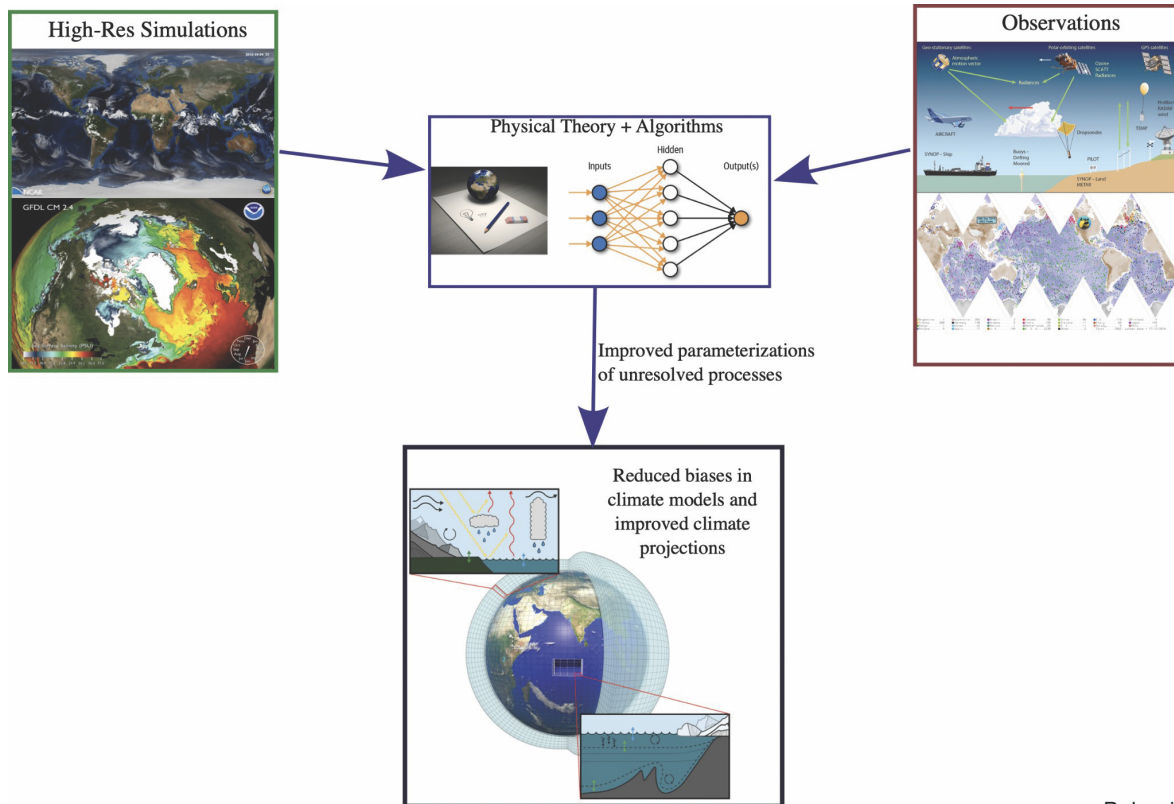
+

increased stratification trapped heat near the surface

ML for ocean model parameterization



ML for ocean model parameterization

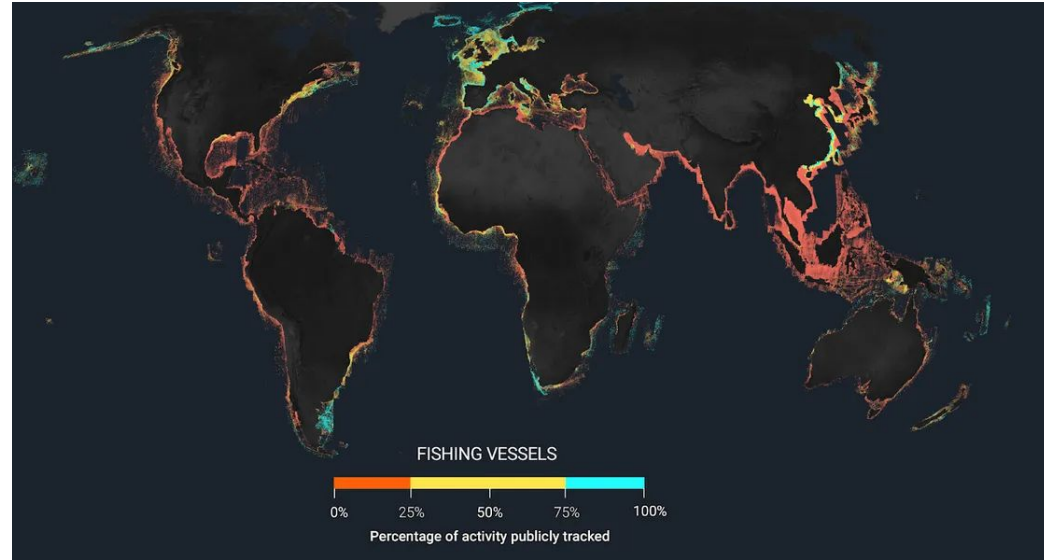


ML for **ocean fisheries management**

Global Fishing Watch provide a near-real-time map of industrial fishing activity to combat illegal activities

They use vessel tracking data and AI

Available data gives e.g. position, speed and heading, but not intent. AI classifies a vessel's *type* (trawler, longliner, purse seiner) and then infer if it is transiting or actively fishing



ML for **ocean fisheries management**:

A note on who benefits

Global Fishing Watch surfaces illegal activity, but enforcement capacity is unevenly distributed

Detection without capacity can shift burden where there is less capacity



MARITIME LAW

Ocean circulation under climate change deep dive

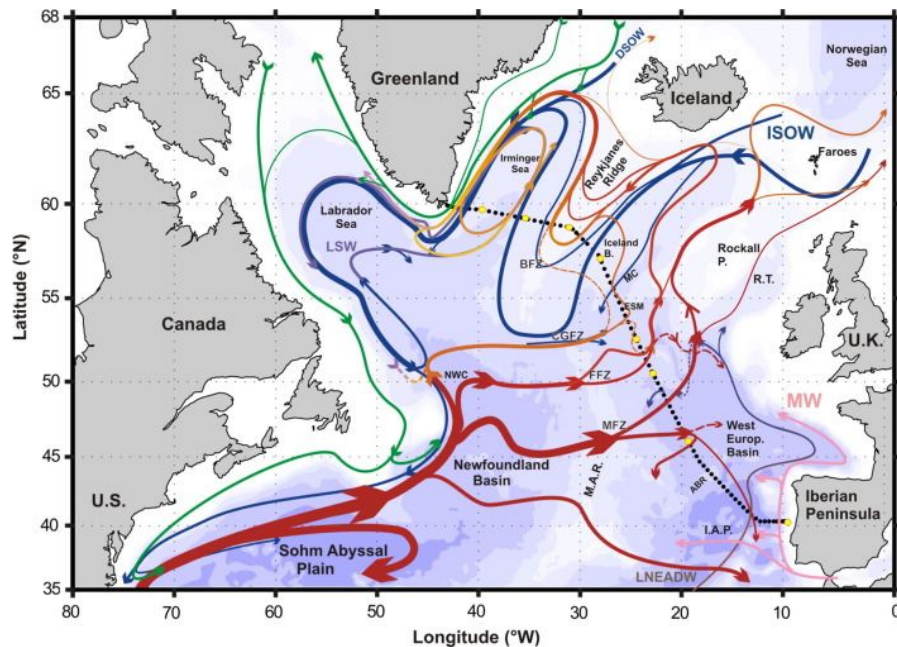
Atlantic Meridional Overturning Circulation (AMOC):

- Heat transport to Europe
- Atmospheric patterns
- Droughts
- Sea level USA East coast
- Hurricanes



Iceland made the move of designating the risk of an AMOC shutdown a **national security threat**

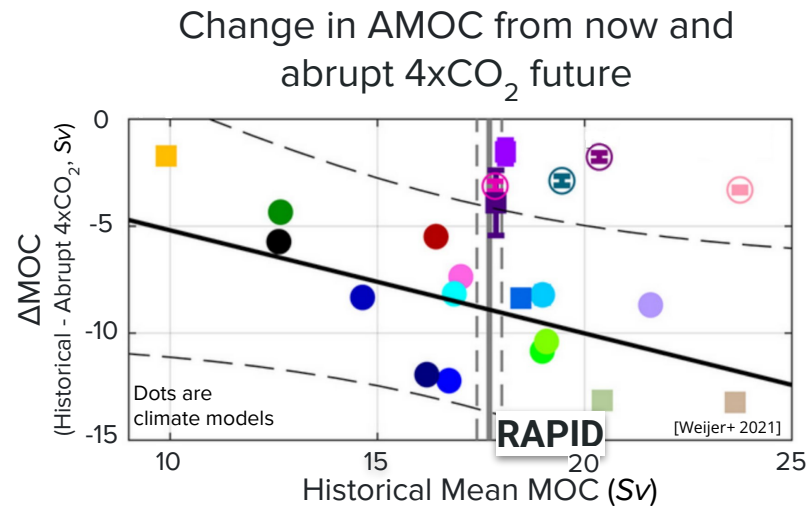
We will use ML to understand the AMOC change in CMIP



Assessed as simplistic metric: Flow rate at 26°N
One number to describe complex currents?

CMIP often has little data to **understand** change

- Bulk metrics common
 - E.g.: Atlantic Meridional Overturning Circulation (A MOC) metric
- Measures the strength and volume transport of the AMOC
- Tracks how fast warm water moves north and cold water returns south
- Current volume transport: Measured using ocean depth, velocity, temperature, and salinity
 - Array Locations: Monitored across latitude lines 26.5°N (RAPID), 34.5°S (SAMBA), and subpolar regions.

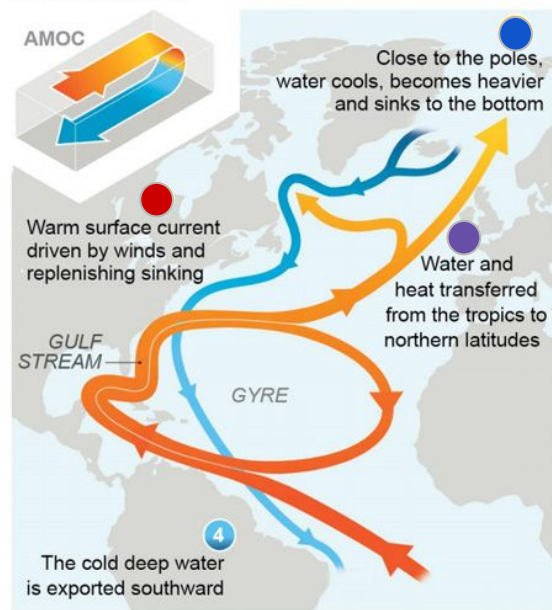


Will the Gulf Stream shut down under climate change?

We can see fingerprints of change in dynamics

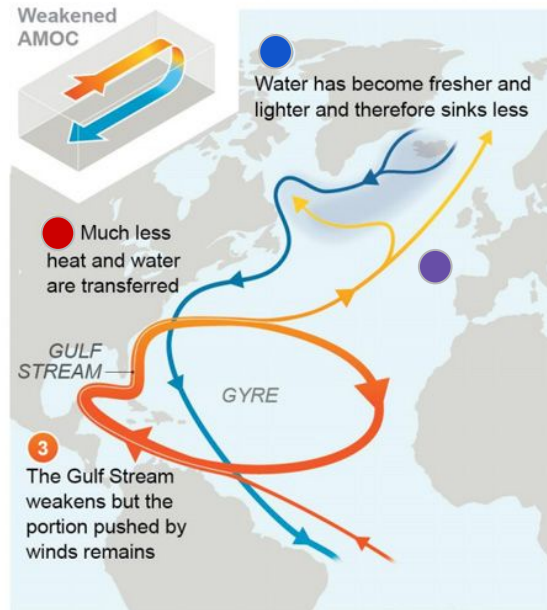
Today

The Gulf Stream is part of both the horizontal, subtropical gyre and the vertical, Atlantic Meridional Overturning Circulation (AMOC)



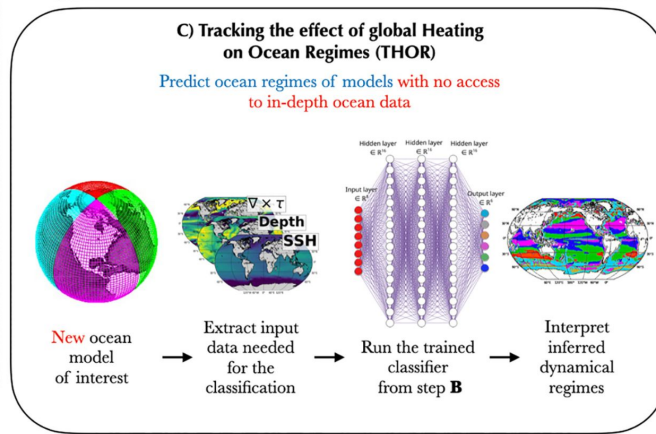
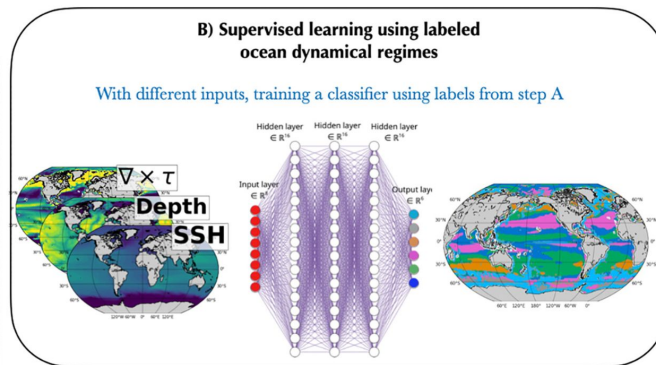
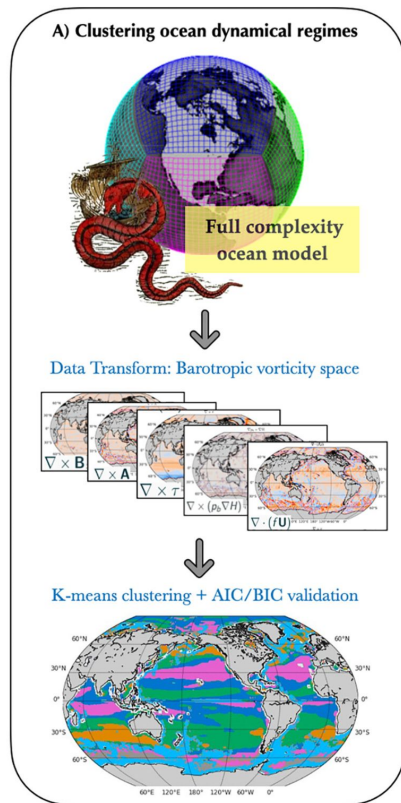
In a warmer world

Climate change weakens the AMOC, which slows the Gulf Stream down



- Gulf stream shifts north
- Path of heat delivery moves
- Less dense/heavy waters

Method: Tracking global Heating with Ocean Regimes (THOR)



Infer current detail with sparse available data

Equitable computing:
open-source and cloud HPC

GitHub [maikejulia/DNN4Cli](https://github.com/maikejulia/DNN4Cli)



Sonnevald and Lguensat, 2021, **JAMES**
Sonnevald, et al., 2021, *International Conference on Machine Learning (ICML) workshop paper spotlight*
Gille [...], Sonnevald et al., 2018, *Open Code Policy for NASA Space Science, white paper*

Tackle overwhelming complexity of fluid dynamics in climate models

Full complexity ocean model component



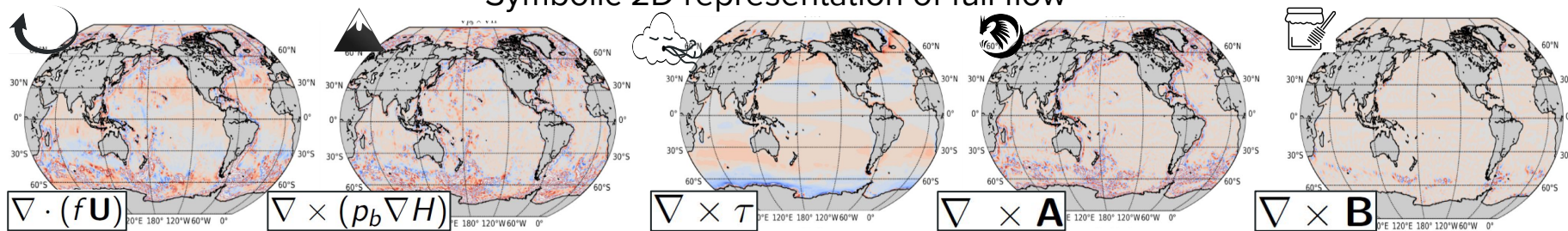
Symbolic Transform



Fundamental equation space framework
(Barotropic vorticity equation)

$$0 = \underbrace{\nabla \cdot (f\mathbf{U})}_{\text{Advection}} - \underbrace{\nabla \times (p_b \nabla H)}_{\text{Wind and Bottom Stress}} + \underbrace{\nabla \times \tau}_{\text{Bottom Pressure Torque}} - \underbrace{\nabla \times \mathbf{A}}_{\text{Lateral Viscosity}} + \underbrace{\nabla \times \mathbf{B}}_{\text{Non-linear Torque}}$$

Symbolic 2D representation of full flow



Tackle overwhelming complexity of fluid dynamics in climate models

Full complexity ocean model component



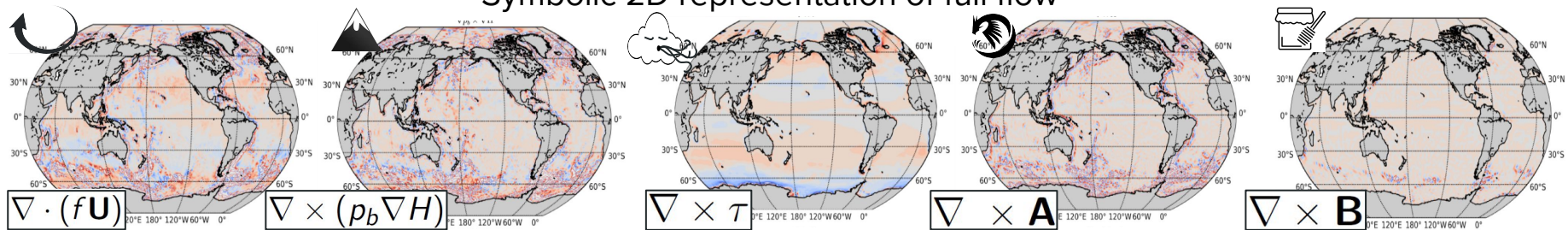
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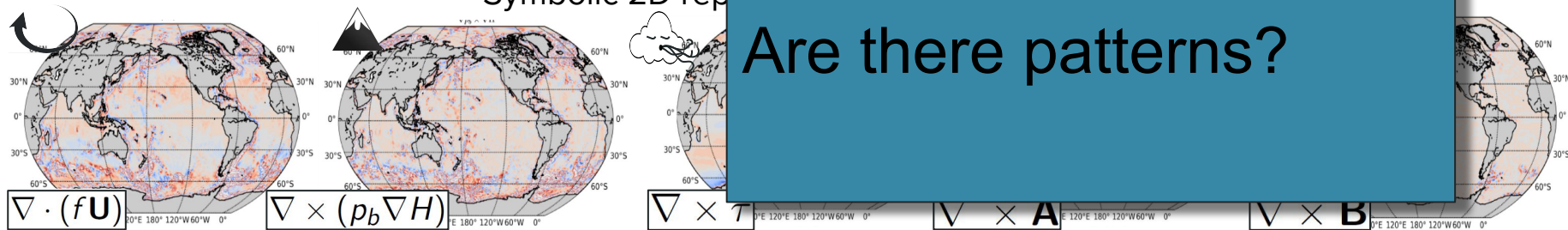


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Symbolic 2D rep

Are there patterns?



Tackle overwhelming complexity of fluid dynamics in climate models

Full complexity ocean model component



Symbolic Transform



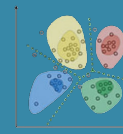
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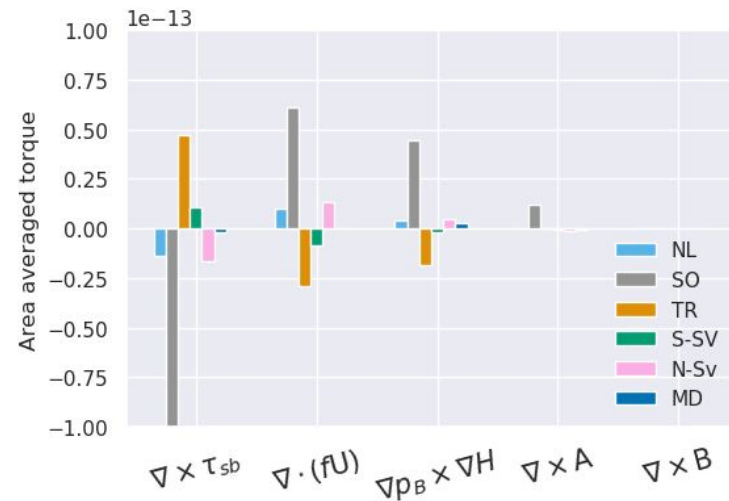
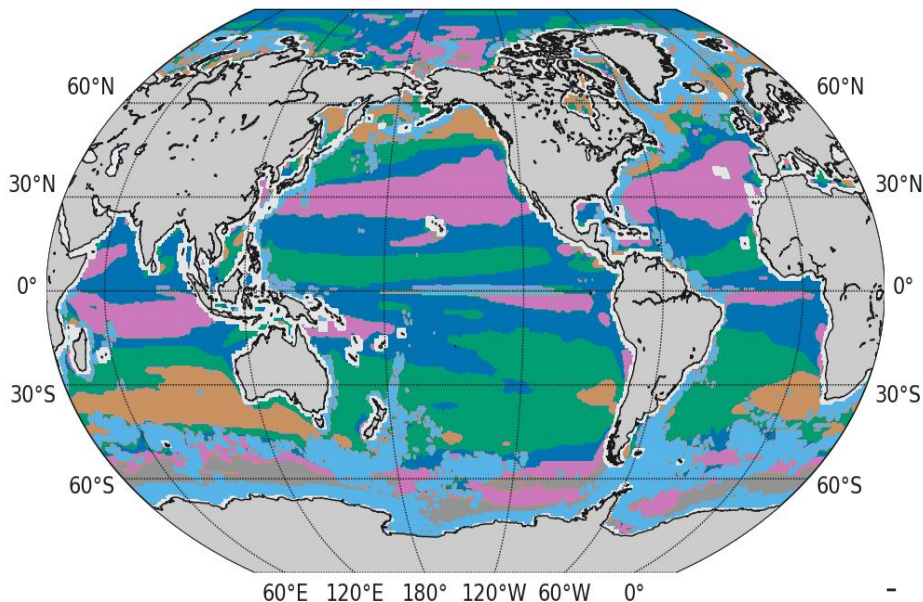
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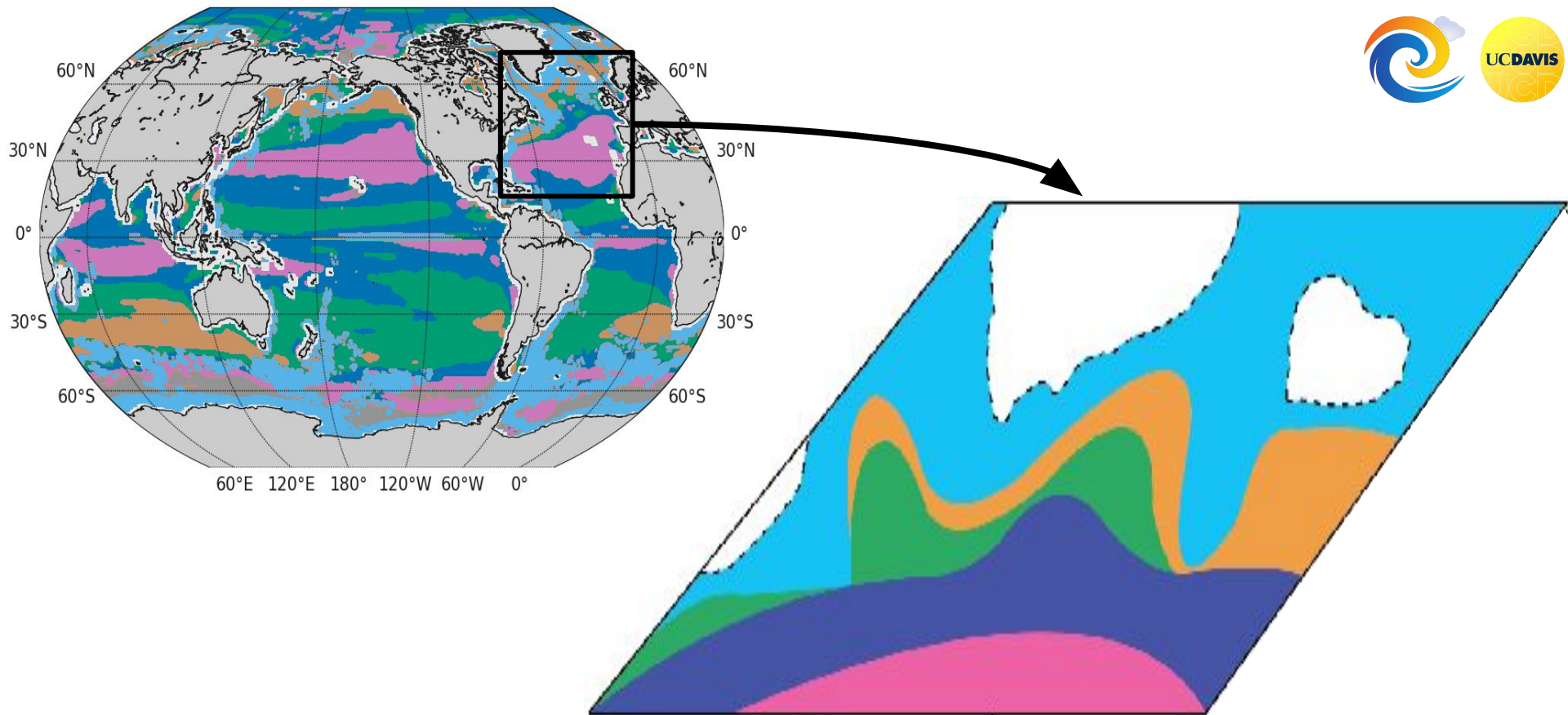
Are there patterns?
Use clustering





- Coherent regions emerge
- Regimes signify balance of terms in the equation

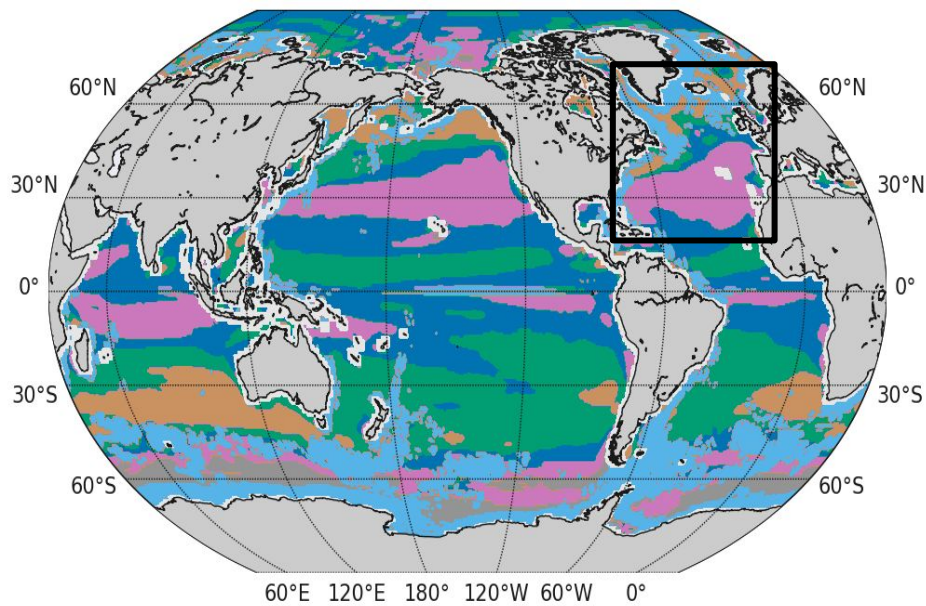
NL-Non-linear	SO- Southern Ocean	TR- Transitional
S-SV- SH Sverdrupian	N-SV- NH Sverdrupian	MD- Momentum dominated



New knowledge: Sonnewald, Reeve & Lguensat, 2023
 Main: Sonnewald, Wunsch & Heimbach, 2019,
 Formalism: Kaiser, Sanches, Sonnewald & Livrenco, 2022

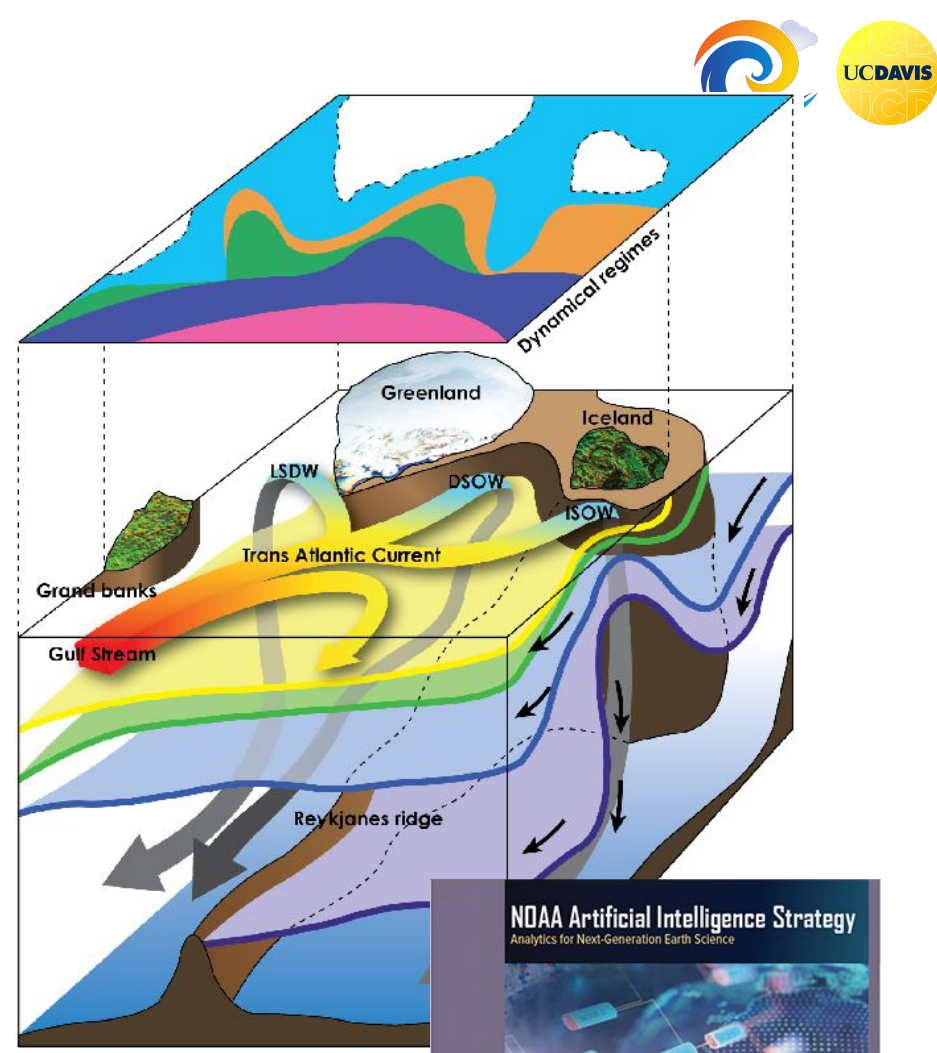
NOAA Artificial Intelligence Strategy
 Analytics for Next-Generation Earth Science



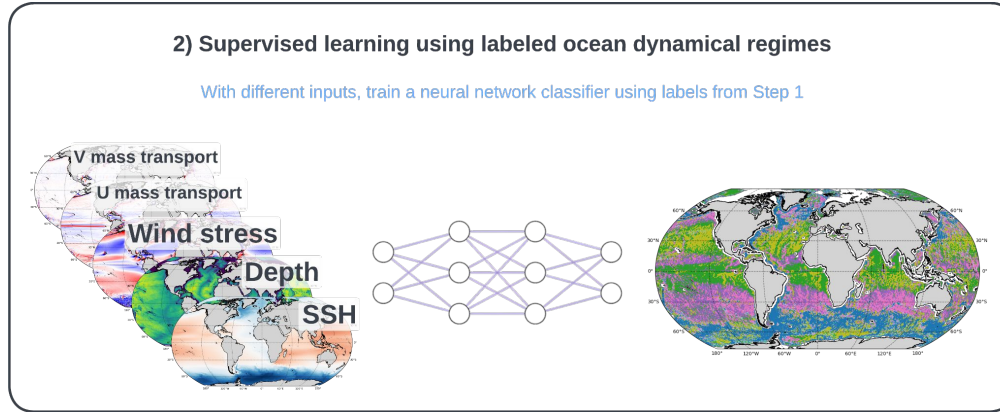


Regimes reveal 3D ocean features tied to e.g. heat transport

- Warm currents brought north (red)
- Cool as basin is crossed (yellow)
- Sink to depth (gray)



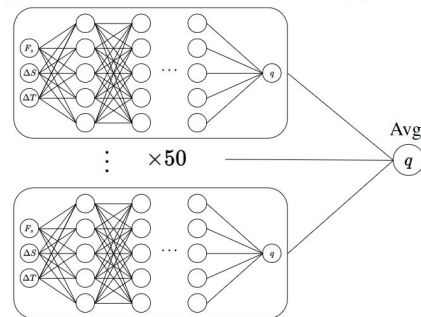
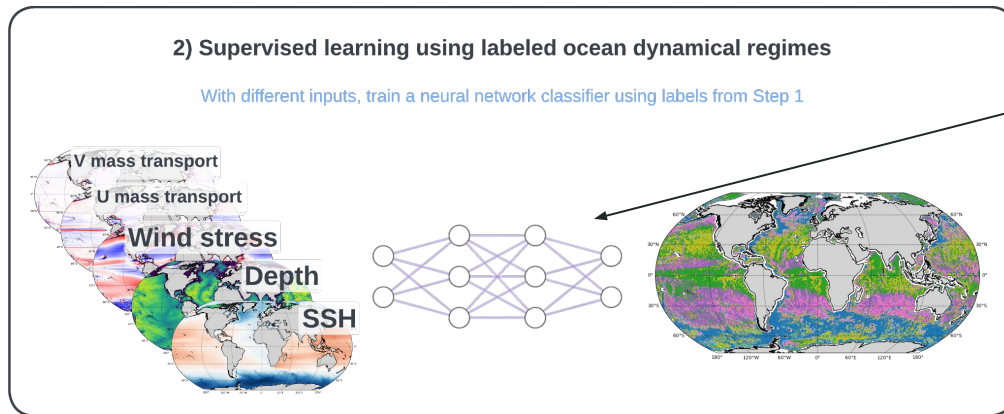
Learning regimes from readily available fields



- Inputs: Approximate “words”
 - Sea surface height (ZOS) + x/y gradients
 - Depth (column height) + x/y gradients
 - Wind stress curl ($\nabla \times \tau$)
 - Depth summed monthly mass transport (umo_2d + vmo_2d)
 - Coriolis parameter (f)
- Labels provide “context”: 6 dynamical regimes
- Deep Ensemble (50 MLP) for uncertainty quantification
- Application of eXplainable AI techniques

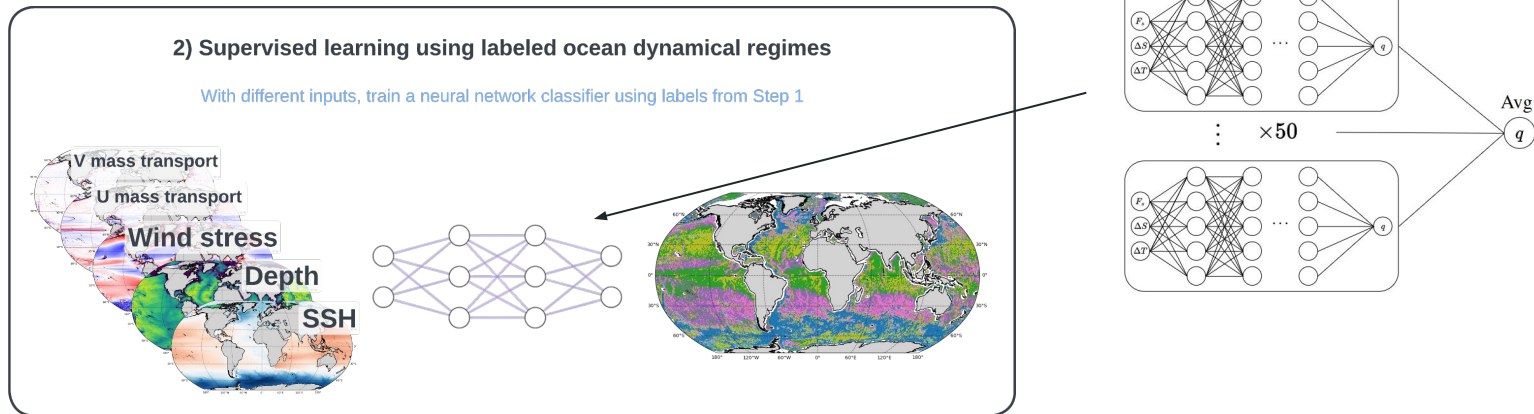
Sonnevald, Reeve & Lguensat, in press, **Nature Comm. Earth and Env.**
Main: **Sonnevald**, Wunsch & Heimbach, 2019, **ESS.**
Formalism: Kaiser, Sanches, **Sonnevald** & Livrenco, 2022 **Eng Appl AI**

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Learning regimes from readily available fields



- Inputs available from CMIP:

- Sea surface height (ZOS)
- Depth (column height)
- Wind stress curl ($\nabla \times \tau$)
- Depth summed monthly mass transport
- Coriolis parameter (f)

- ML model:

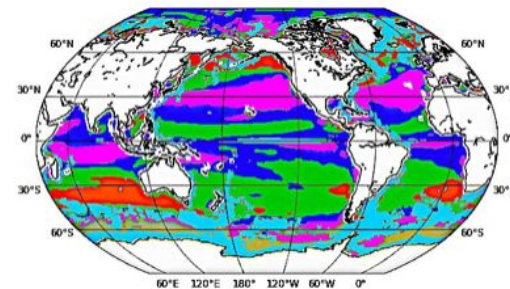
- Deep Ensemble (50 MLP) for uncertainty quantification
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Apply to different climate scenarios to understand changes

THOR: Recognizing regimes in an unseen model

Classification

Predict dynamical regimes in unseen data

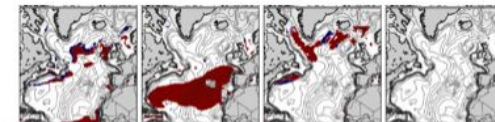


Mean of predictions is used to find the most probable class

XAI method (ex: LRP) is used for each MLP to obtain the mean of relevances

Explainability

Attribute predictive skill to input features

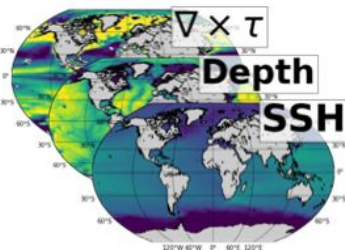


Negative relevance 0 Positive relevance

Ensemble of trained MLP

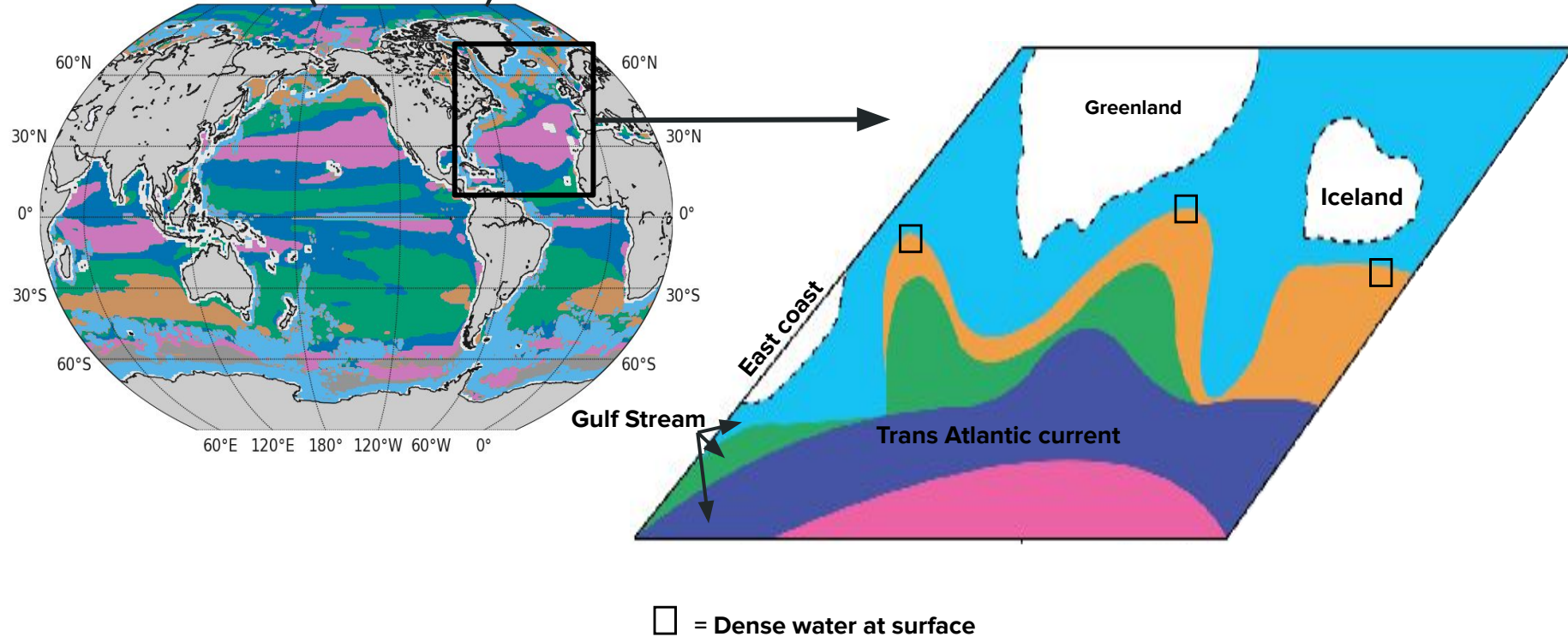


Ocean model of interest

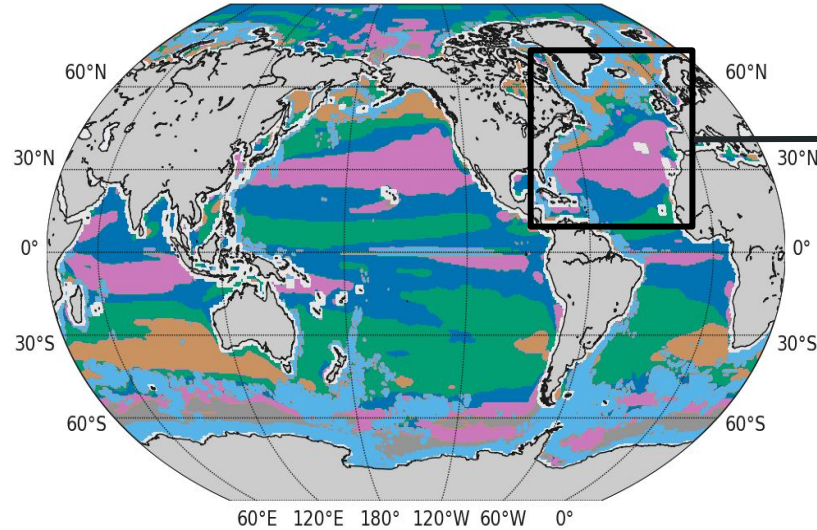


Extract input data needed by the classifier

THOR for diagnosing processes of the Atlantic meridional overturning circulation (AMOC)



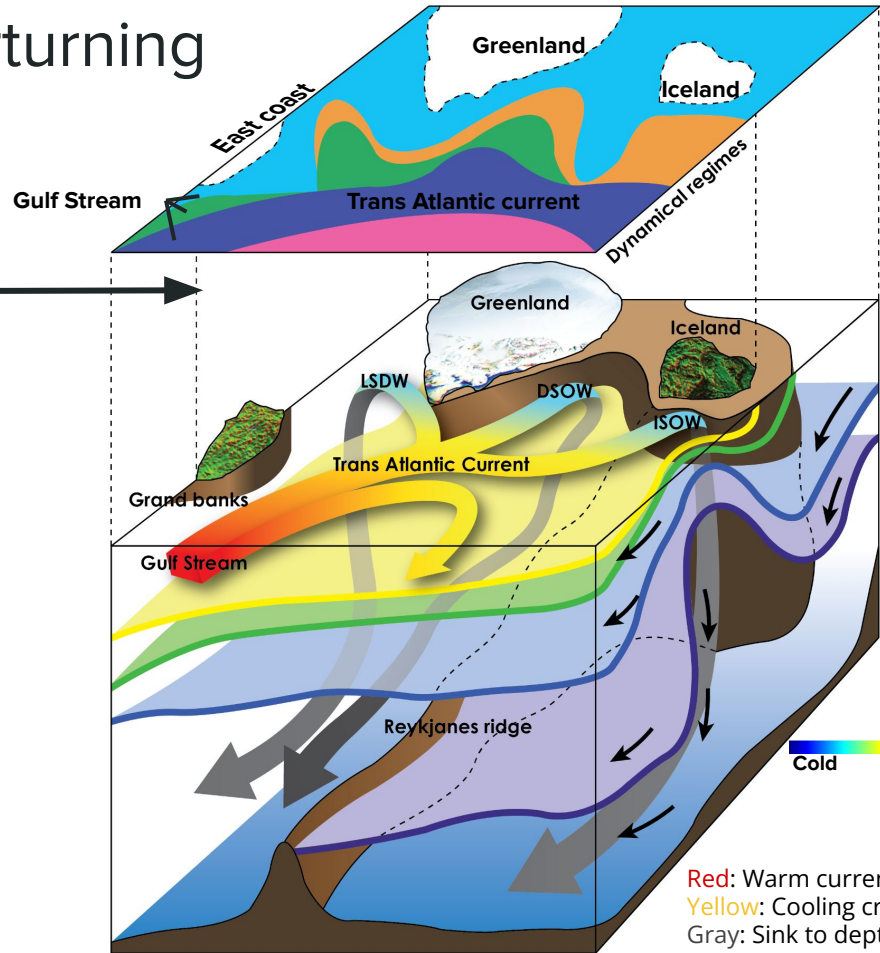
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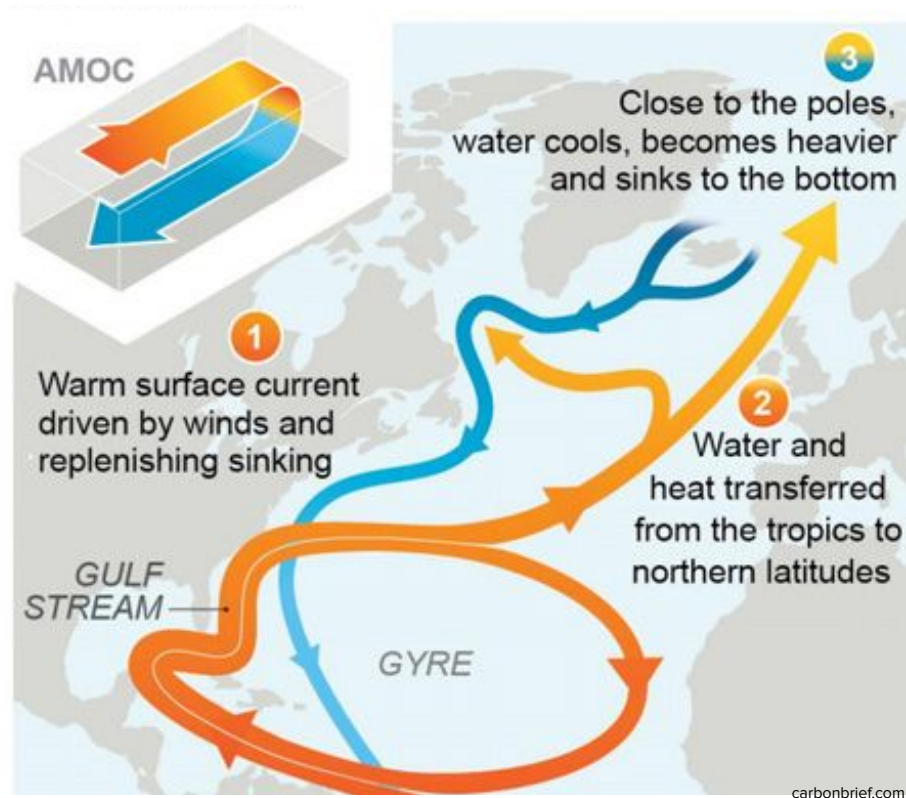
THOR used for CMIP models

Diagnose processes to speed up tuning and parameterisation of models

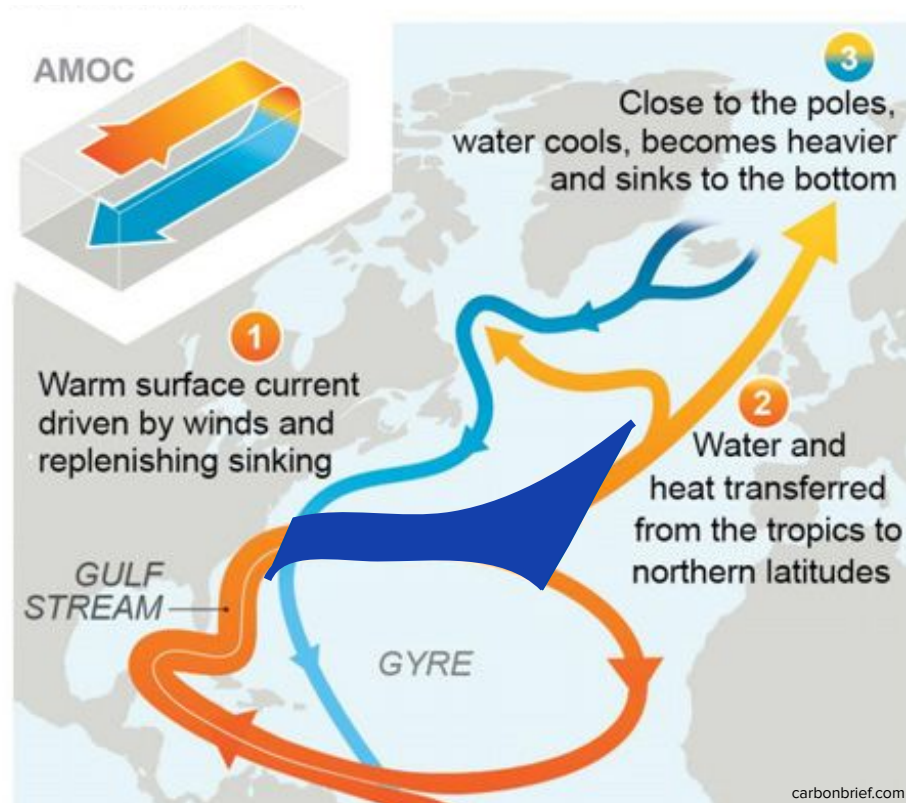
GFDL and IPDL



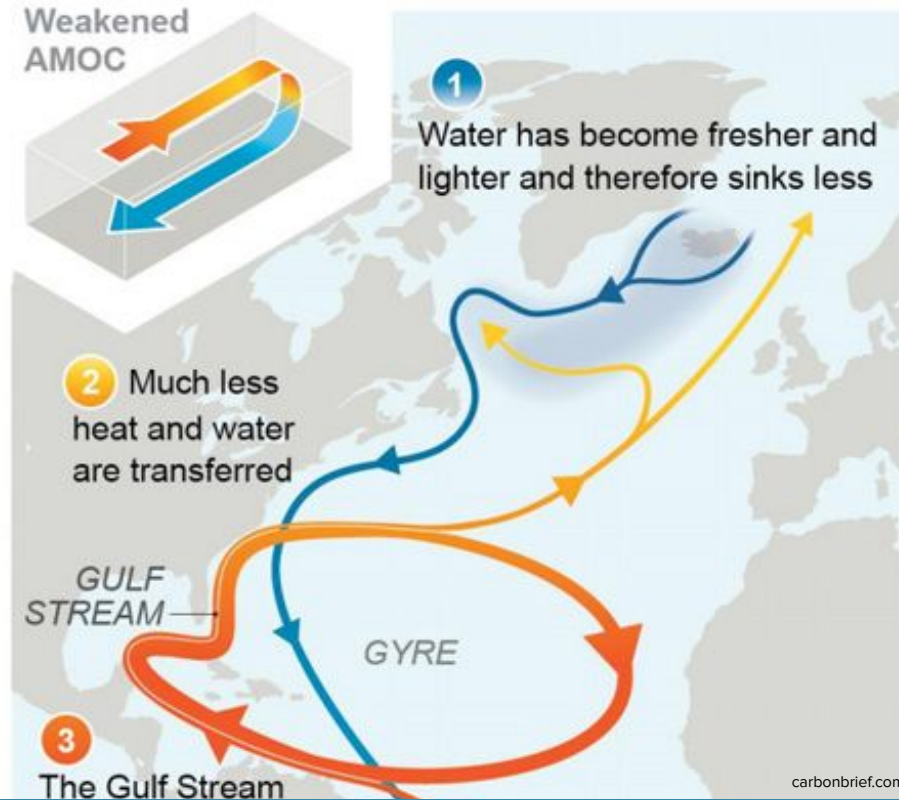
How will the overturning weaken under climate change?



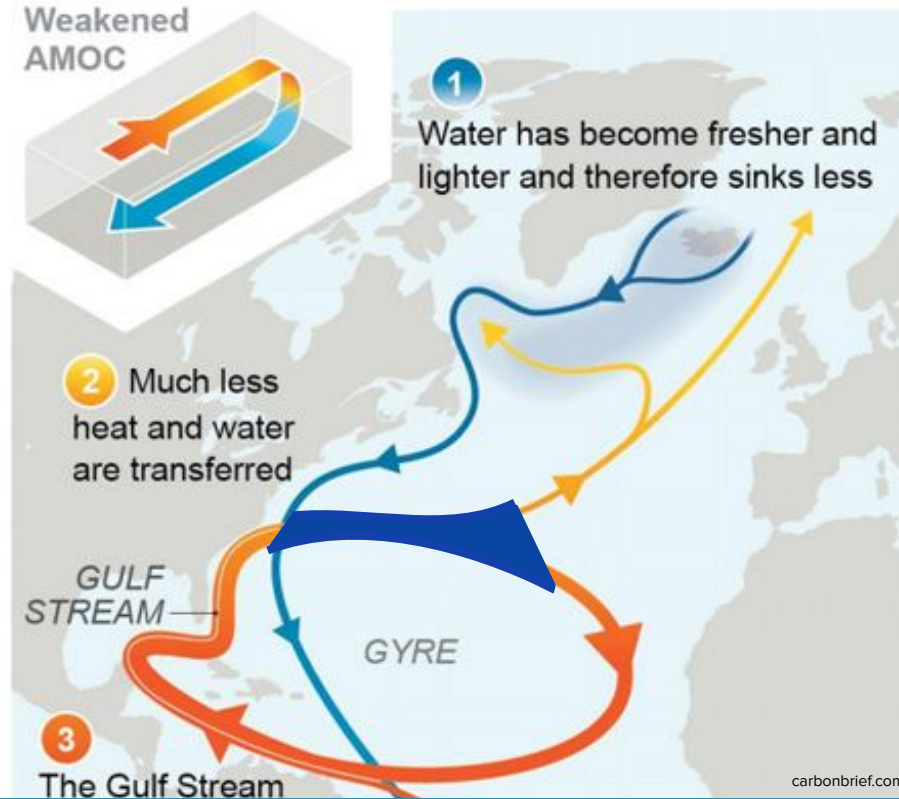
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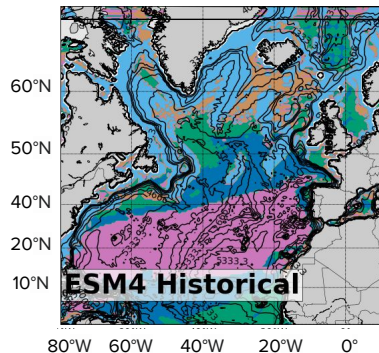
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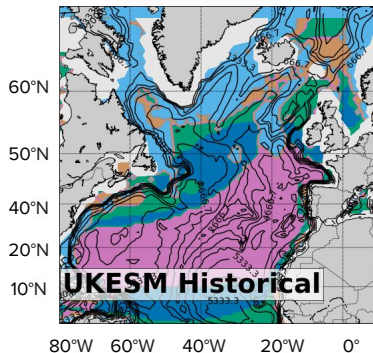
THOR reveals processes changes in abrupt CO₂ quadrupling scenario

'Historical' scenarios look similar but future scenarios reveal disparities

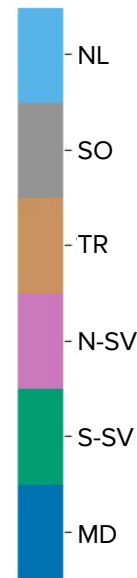
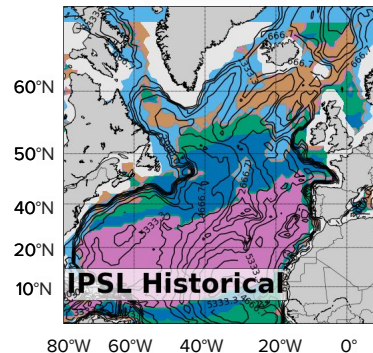
GFDL ESM4 



UKESM 



IPSL 



UKESM: [Senior+ 2020]
IPSL: [Boucher+ 2020]

THOR reveals processes changes in abrupt CO₂ quadrupling scenario

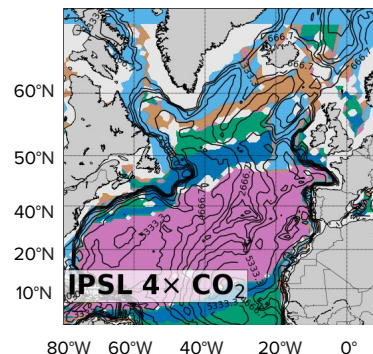
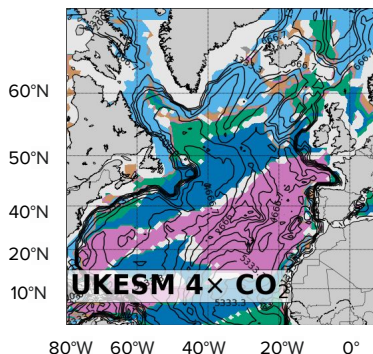
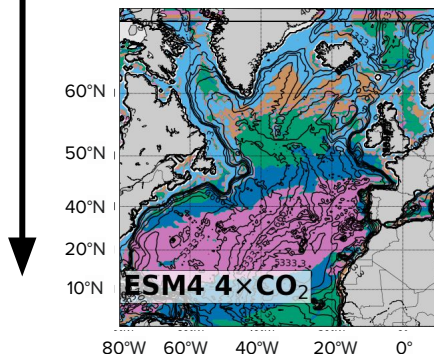
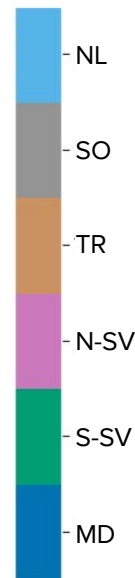
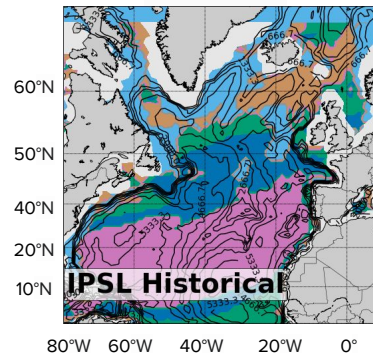
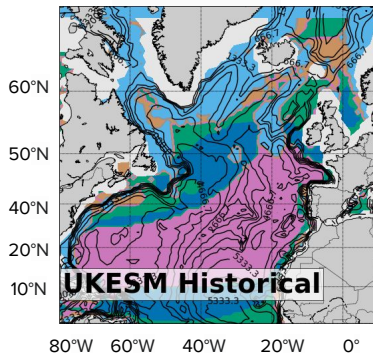
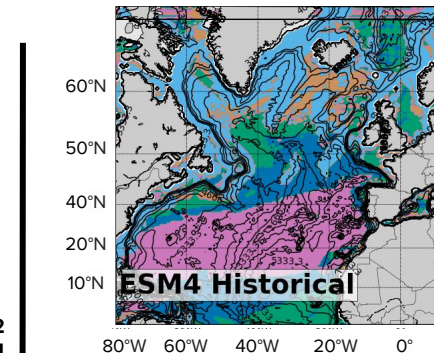
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GFDL ESM4 

UKESM 

IPSL 

CO₂
forcing



THOR reveals processes changes in abrupt CO₂ quadrupling scenario

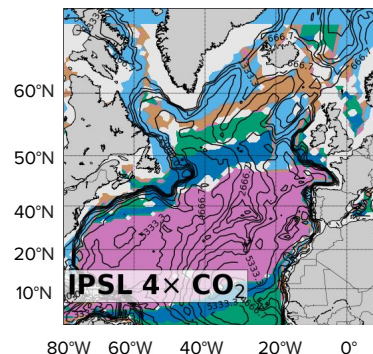
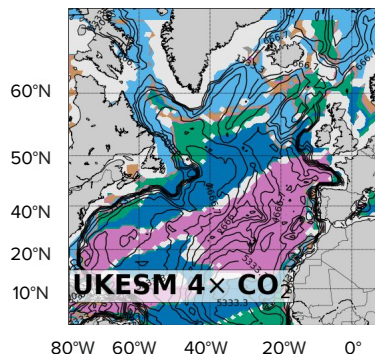
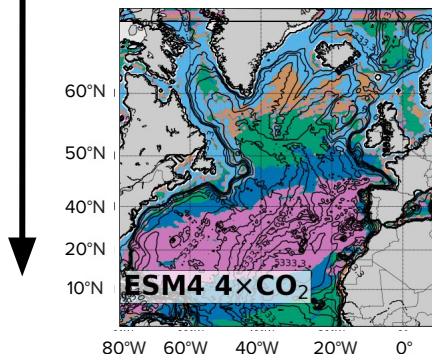
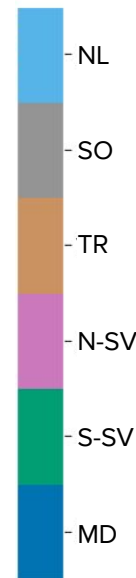
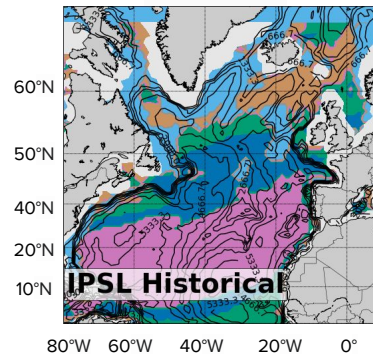
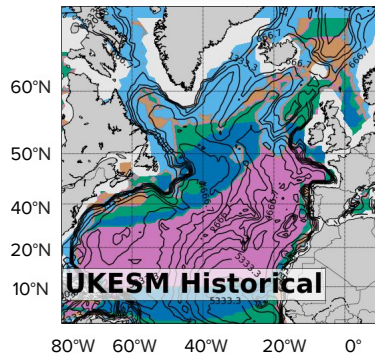
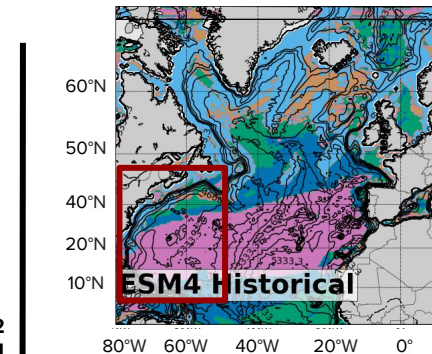
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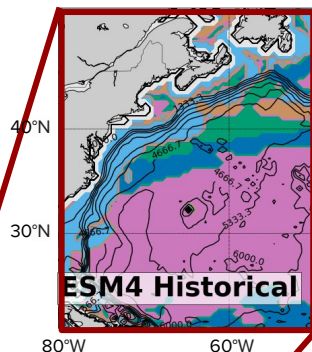
CO₂
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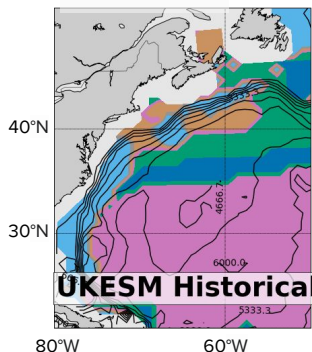
THOR reveals processes changes in abrupt CO₂ quadrupling scenario

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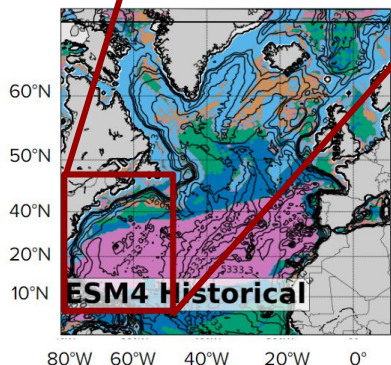
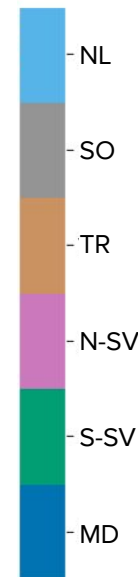
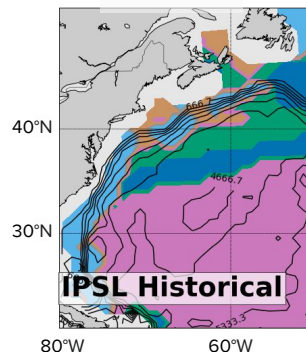
GFDL ESM4 



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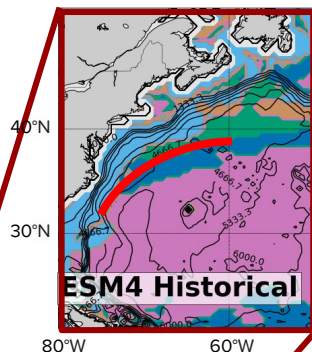


UKESM: [Senior+ 2020]
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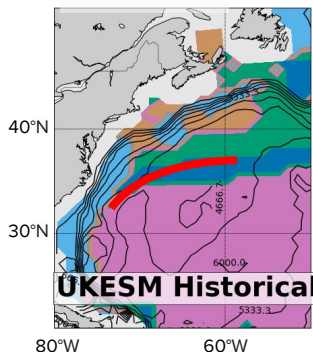
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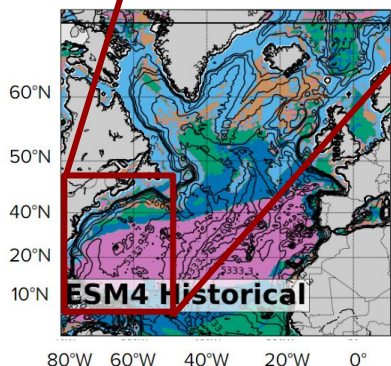
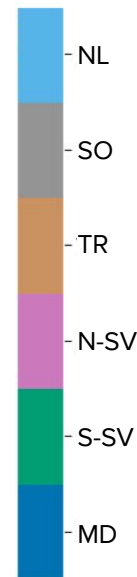
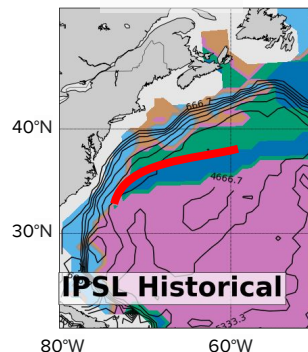
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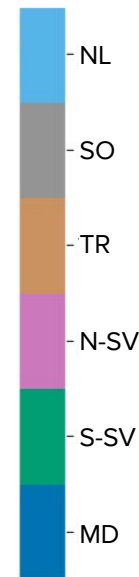
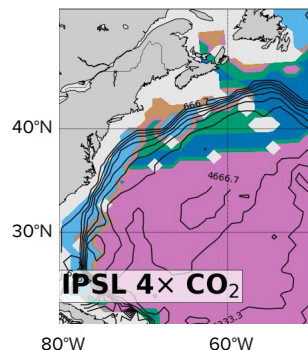
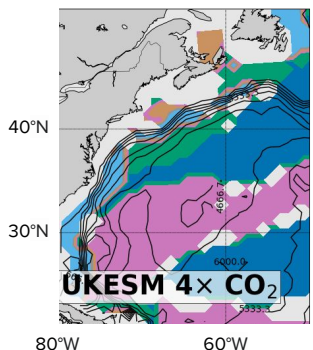
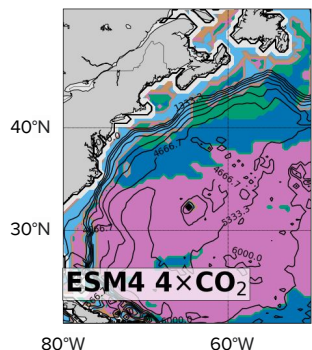
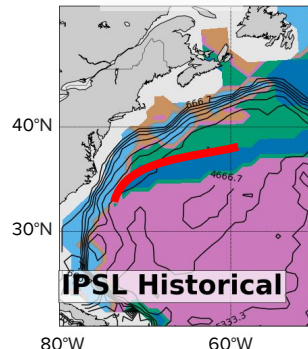
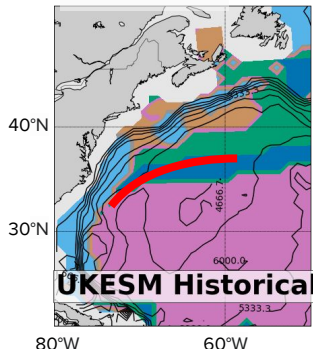
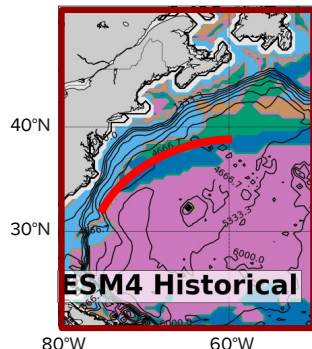
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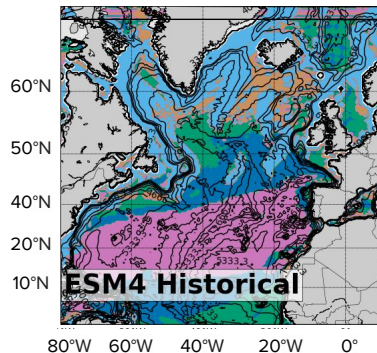


CO₂ forcing

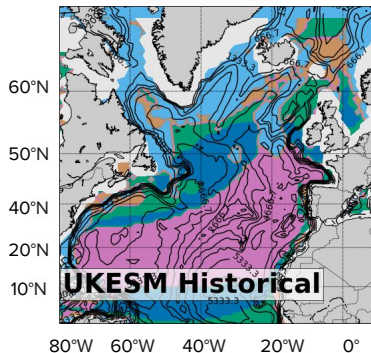
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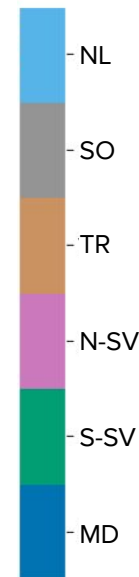
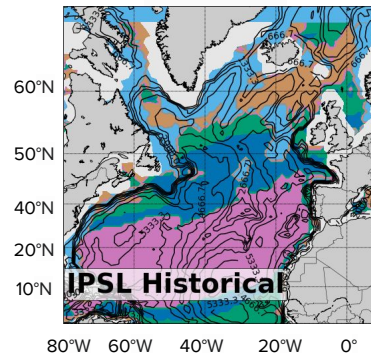
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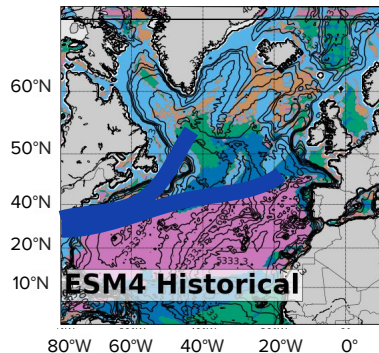


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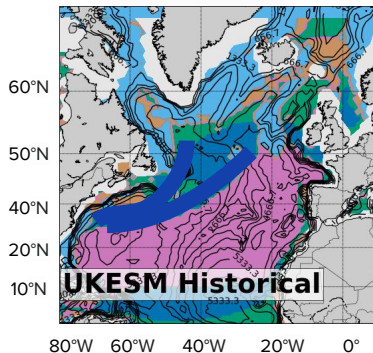
THOR reveals processes changes in abrupt CO₂ quadrupling scenario

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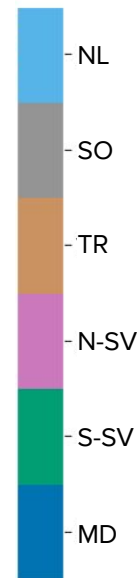
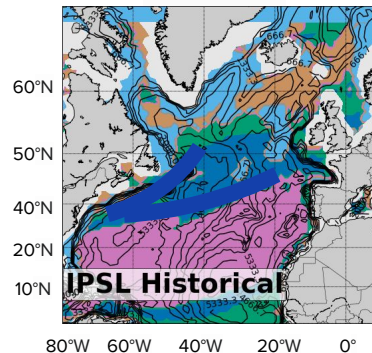
GFDL ESM4 



UKESM 



IPSL 



UKESM: [Senior+ 2020]
IPSL: [Boucher+ 2020]

THOR reveals processes changes in abrupt CO₂ quadrupling scenario

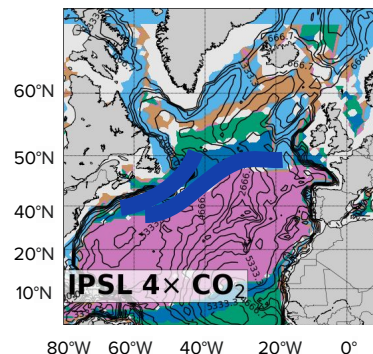
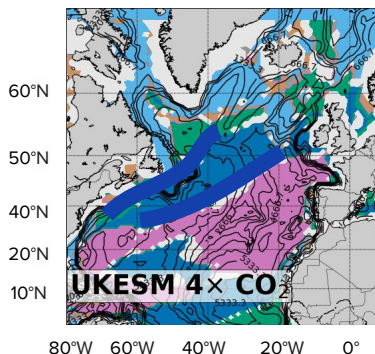
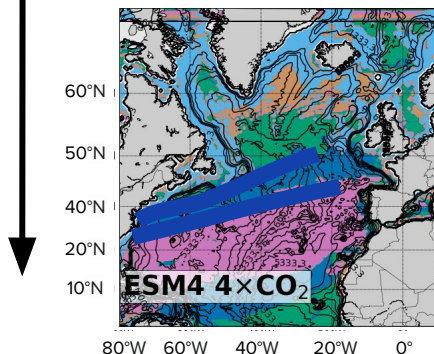
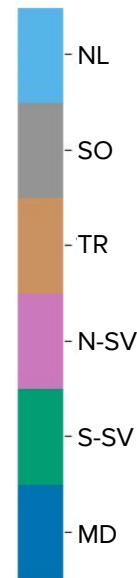
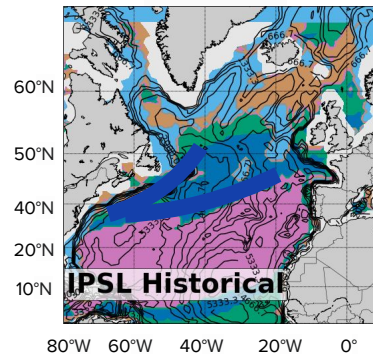
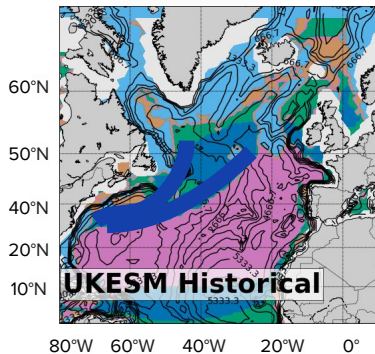
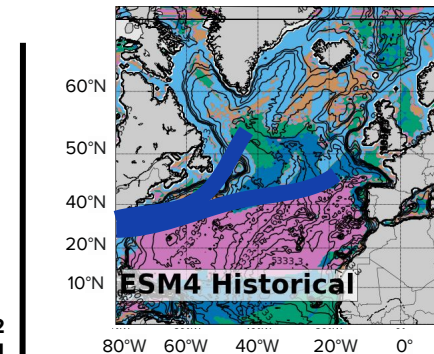
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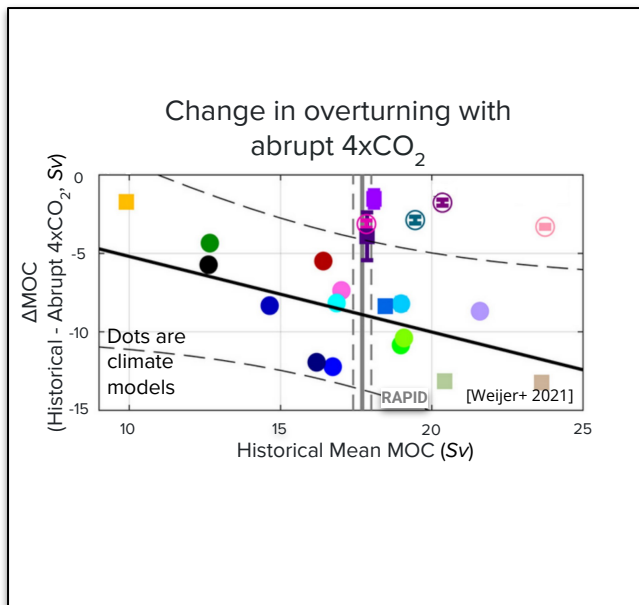
IPSL 

CO₂
forcing

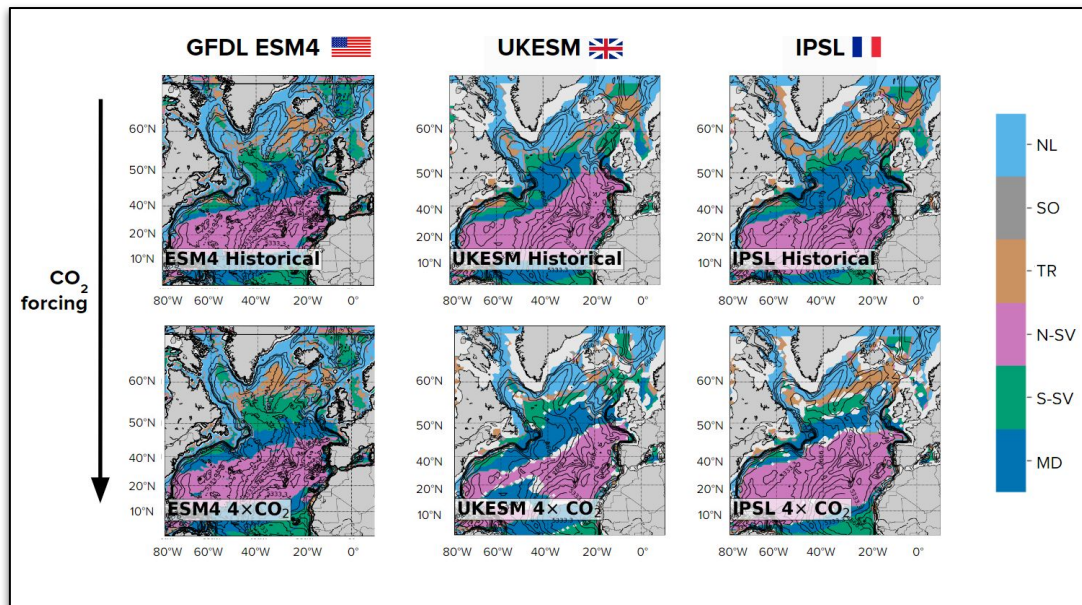


New method offers great improvement

Previous



New method from **SL 2021 JAMES**



Datasets

Physical ocean:

Global Ocean Observing Systems (GOOS): <https://www.ocean-ops.org/goosreport/>

Copernicus Marine (CMEMS) for reanalysis, satellite and forecasts data: <https://marine.copernicus.eu/>

CMIP6 Climate model data: <https://esgf.github.io/>

Ocean Biogeochemistry and ecosystems:

Surface Ocean CO₂ Atlas (SOCAT): <https://socat.info/>

NASA OceanColor, and PAC: <https://pace.gsfc.nasa.gov/>

Marine biodiversity data Ocean Biodiversity Information Systems (OBIS): <https://obis.org/>

Species occurrence Global Biodiversity Information Facility (GBIF): <https://www.gbif.org/>

Human pressures:

Global Fishing Watch (open API): <https://globalfishingwatch.org/>

Allen Coral Atlas: <https://www.allencoralatlas.org/>

Polar: MEOP: <http://www.meop.net/> (the seals!)

Tools

Ocean AI model Samudra: <https://m2lines.github.io/Samudra/docs/>

Using the data Python toolbox Pangeo stack: <https://pangeo.io/>

Resource for compute:

Google Colab: <https://colab.research.google.com/>

Copernicus WEkEO: <https://wekeo.copernicus.eu/>

Exciting (big!) open problems

Coastal and shelf seas where models and satellites are worst:

Complex bathymetry, river inputs, and tides made this difficult. Downscaling and emulation for coastal risk.

Extrapolation to future climate states:

A model trained on the historical ocean asked about $4\times\text{CO}_2$ is extrapolating by construction. THOR and OceanCBM are work on this.

Sub-grid parameterisations in models:

ML schemes that look good offline and then drift or blow up online. Very active field.

Interpretability that is mechanistically valid:

OceanCBM is one attempt, the general problem is very open.

Small-scale fisheries:

Detection that doesn't depend on cooperative broadcast.

Multi-modal fusion:

Ocean colour + in situ + acoustics + eDNA for community composition.

How to get involved!

Researchers and students:

Come to meetings: Ocean Sciences Meeting, EGU/AGU, Climate Informatics, and the CCAI workshops at NeurIPS/ICML/ICLR

Contribute to codes e.g. [THOR](#) (regime approach transfers to other basins and variables) or Samundra. There are many!

NGOs and governments:

The UN Ocean Decade endorsement route, GOOS and its regional alliances (LOGOOS, GOOS Africa), IOC-UNESCO's ODIS, and Global Fishing Watch's partnership programme

Funders:

Funding for ocean climate work is cut in areas traditionally supporting. While all research is important, especially sustained time series, observation infrastructure in the Global South, and *capacity* should be supported

AI for Oceans & Marine Systems

The ocean in the climate system

97% of the water, 90% of the excess heat, ~25% of our CO₂, >50% of the oxygen

Simple physics, with emergent complex patterns

Density contrasts drive an overturning circulation with global reach

Consequences of a changing climate

Warming, acidification and heatwaves reach primary production, zooplankton and fisheries

Data rich, insight poor or the opposite?

Observations, satellites and models each cover only part of the problem

ML is the bridge, not the destination!

